

Multi-Omic Integration of Monoglyceride Response in Model Adipose Tissue Macrophages with Obesity

Transcriptomics, Lipidomics, and Spatial Lipidomics

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7 May 2025

A thesis in fulfillment of the Masters in Biology Degree

Abstract

Obesity fosters a low-grade, chronic inflammation within adipose tissue, primarily driven by macrophages infiltrating and forming crown-like structures (CLSs) around hypertrophic adipocytes. These macrophages secrete pro-inflammatory cytokines, disrupt insulin signaling, and remodel local lipid pools, reinforcing metabolic dysfunction. To elucidate how specific glycerolipids and fatty-acid-derived mediators shape macrophage behavior, we opted for a three-pronged, multi-omics framework, integrating spatial lipidomics (MALDI IMS), bulk lipidomics (LC-MS), and transcriptomics (RNA-seq) in obese adipose macrophages. First, spatial lipidomics via MALDI IMS enabled high-resolution mapping of monoglyceride (MG) and triglyceride (TG) “hotspots” in tissue sections, pinpointing local accumulations frequently overlapping with macrophage-dense or necrotic areas. This spatial view reveals how lipids concentrate within inflammatory niches, potentially fueling macrophage-driven responses. Next, bulk lipidomics (LC-MS) characterized the global glycerolipid profile, particularly focusing on arachidonic acid (AA) and 2-arachidonoylglycerol (2-AG), two key molecules linked to eicosanoid and endocannabinoid pathways, respectively. Principal component analysis (PCA) and hierarchical clustering indicated that AA induces a pronounced remodeling of macrophage lipid pools, aligning with its pro-inflammatory potential via eicosanoid precursors. Conversely, 2-AG exerted subtler or more targeted effects, consistent with a modulatory rather than a strongly pro-inflammatory role. To anchor these lipid shifts to molecular mechanisms, transcriptomic (RNA-seq) profiling identified differentially expressed genes (DEGs) tied to β -oxidation, eicosanoid/endocannabinoid metabolism, and inflammatory polarization. Notably, AA-treated macrophages often upregulated enzymes like COX2 or ALOX5, whereas 2-AG-treated cells revealed comparatively modest transcriptional alterations, underscoring 2-AG’s less disruptive yet potentially immunoregulatory impact. By correlating local MG/TG distributions from MALDI IMS with total lipid abundances and gene-expression patterns, we gained a holistic view of how glycerolipid metabolism, endocannabinoid signaling, and macrophage phenotype intersect in obese adipose tissue. This multi-omic integration thus provides a robust platform for unraveling pro- vs. anti-inflammatory pathways, paving the way for targeted interventions in metabolic disease.

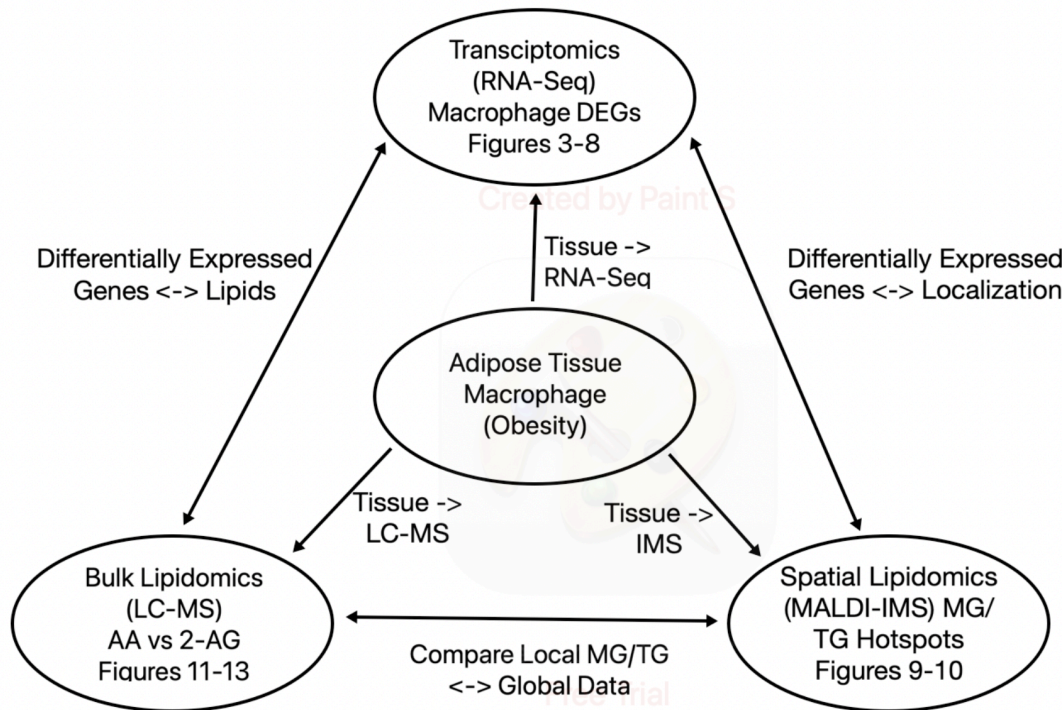


Figure 1: Multi-Omic Integration of Adipose Tissue Macrophages.

This schematic motivated by the findings of Alemán *et al.* (2018) illustrates how three complementary omic workflows: spatial lipidomics (MALDI-IMS; Figs 9–10), bulk lipidomics (LC–MS; AA vs 2-AG; Figs 11–13), and transcriptomics (RNA-Seq; macrophage DEGs; Figs 3–8) are combined to characterize macrophage adaptations within obese adipose tissue. Tissue biopsies feed into each pipeline (“Tissue → IMS,” “Tissue → LC–MS,” “Tissue → RNA-Seq”), while bidirectional arrows overlay the critical correlations:

- **Lipids ↔ DE Genes:** linking global lipid abundance changes to shifts in macrophage gene expression.
- **Genes ↔ Localization:** mapping differentially expressed genes onto spatially resolved MG/TG hotspots.
- **Local MG/TG ↔ Global Data:** comparing localized lipid intensities against whole-tissue lipidome profiles to uncover microenvironment-driven metabolic cues.

Together, these integrated layers illuminate how localized and systemic lipid signatures coordinate with gene regulation to drive obesity-associated macrophage phenotypes.

Introduction

Obesity, Adipose Tissue, and Chronic Inflammation

Obesity and its associated metabolic syndromes represent a global health challenge, characterized by excess adiposity, chronic inflammation, and insulin resistance. Over the past few decades, obesity rates have soared worldwide, significantly raising the prevalence of type 2 diabetes, cardiovascular disease, and other comorbidities. Adipose tissue, once viewed as a passive energy reservoir, is now recognized as an active endocrine organ that secretes a wide variety of adipokines (ex. leptin, adiponectin) and inflammatory cytokines (TNF- α , IL-6, IL-1 β). Through these secretions, adipose tissue exerts far-reaching influences on local and systemic metabolism, making adipose tissue a central hub of metabolic regulation [1–3]. Within this tissue, macrophages play a pivotal immune role, affecting processes from tissue remodeling and immune surveillance to inflammatory responses. Obesity is widely defined by an excessive accumulation of adipose tissue that predisposes individuals to numerous metabolic dysfunctions. Under lean conditions, adipose tissue operates with minimal inflammatory signals, maintaining homeostasis. However, excess nutrient intake leads to adipocyte hypertrophy and hyperplasia, triggering the secretion of chemokines and free fatty acids that recruit immune cells particularly macrophages into the tissue [4]. The persistent inflammatory milieu contributes to insulin resistance and fosters a cycle wherein heightened inflammation further disrupts metabolic equilibrium [5]. Understanding the molecular cues behind these macrophage-driven processes is thus critical for managing obesity-related diseases.

Macrophage Polarization in Obese Adipose Depots

Within adipose tissue, macrophages can adopt distinct functional states. In lean adipose tissue, macrophages often exhibit an M2-like (anti-inflammatory) phenotype, aiding in tissue remodeling and debris clearance. However, as obesity progresses, macrophages shift toward an M1-like (pro-inflammatory) phenotype, characterized by elevated secretion of TNF- α , IL-6, and other cytokines [6]. Histologically, these M1 macrophages tend to cluster around necrotic adipocytes, forming “crown-like structures” that signify local hotbeds of inflammation [7]. This process exacerbates insulin resistance and systemic metabolic dysfunction. In lean individuals, the relatively small size of adipocytes and limited release of pro-inflammatory mediators help maintain an M2-dominant macrophage landscape, promoting efficient lipid uptake and tissue homeostasis [1,2]. As excess weight gain ensues,

however, adipocytes undergo hypertrophy and hyperplasia, secreting an array of signals—including free fatty acids, chemokines, and cytokines—that stimulate macrophage infiltration [3,4]. Over time, this influx of immune cells amplifies local inflammation, pushing resident macrophages toward an M1-like phenotype. The characteristic “crown-like structures” observed around dying or stressed adipocytes reflect this proinflammatory niche, further fueling metabolic dysregulation and insulin resistance [5]. Ultimately, the continual feedback of adipocyte-derived factors, coupled with M1 macrophage cytokine release (ex. TNF- α , IL-1 β , IL-6), fosters a self-reinforcing cycle of chronic inflammation that worsens metabolic outcomes. Understanding how macrophages transition from an M2- to M1-like state—and how this transition could be modulated—is central to developing therapeutic interventions that mitigate obesity’s detrimental effects on adipose tissue function.

Figure 2 (below) illustrates the transition from lean, M2-predominant adipose tissue to an inflamed, M1-dominated environment with crown-like structures. Notably, free fatty acids released by enlarged adipocytes further amplify macrophage infiltration, sustaining a chronic inflammatory loop [8].

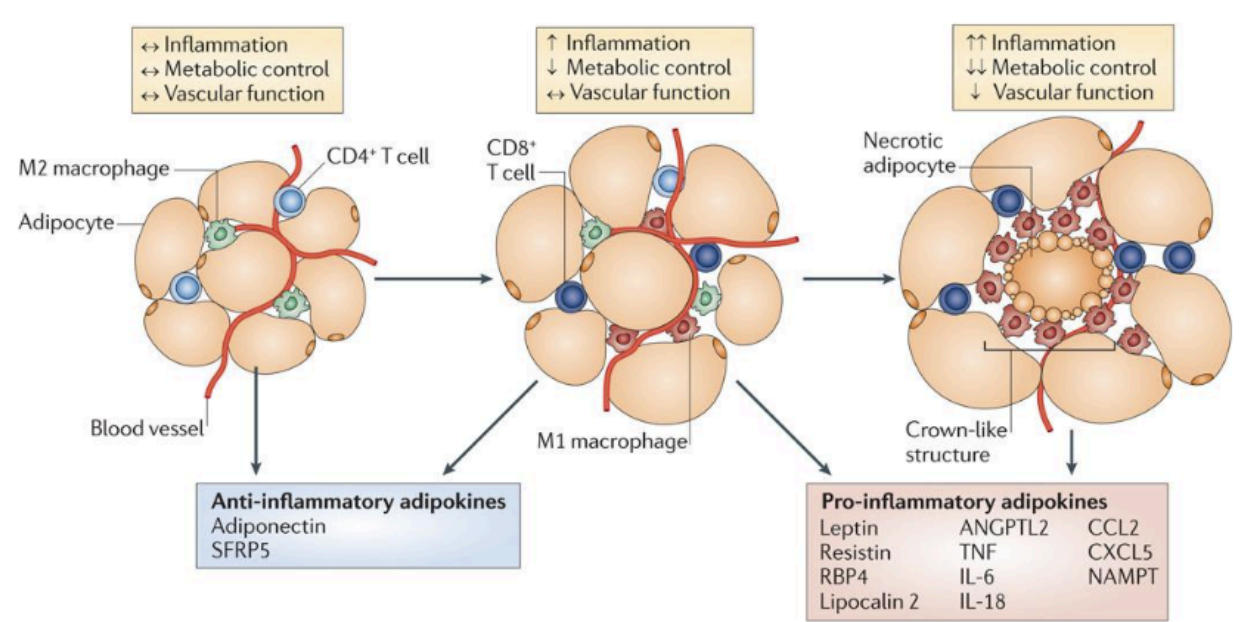


Figure 2: Progression of adipose tissue macrophage polarization under lean and obese states. In the lean condition (left), macrophages predominantly exhibit an anti-inflammatory “M2-like” phenotype, helping maintain metabolic homeostasis. As adipocytes expand (center), heightened lipid and cytokine release promotes a shift toward a pro-inflammatory “M1-like” phenotype. Ultimately (right), macrophages form

crown-like structures around necrotic adipocytes and secrete cytokines (TNF- α , IL-6, IL-18), perpetuating low-grade chronic inflammation and metabolic dysregulation characteristic of obesity.

Monoglycerides: Energy Substrates and Signaling Molecules

While free fatty acids (FFAs) and triglycerides have long been implicated in macrophage biology, monoglycerides (MAGs)—molecules consisting of a single fatty acid chain bound to a glycerol backbone—have recently gained attention for their multifaceted roles [9,10]. Evidence suggests MAGs can serve as:

1. **Energy Substrates:** Certain monoglycerides can undergo hydrolysis to release FFAs, which macrophages may oxidize via β -oxidation for ATP production [11]. This route may bolster a more oxidative, potentially less inflammatory macrophage phenotype.
2. **Signaling Mediators:** MAGs like 2-arachidonoylglycerol can engage endocannabinoid receptors (ex. CB2, GPR119), influencing inflammation and lipid metabolism [12]. Variations in the fatty acyl chain can alter their signaling potency, adding complexity to how MAGs modulate immune responses.

However, MAG metabolism in obese adipose tissue remains poorly characterized. Some studies indicate excess MAGs can exacerbate inflammation under certain conditions, fueling a shift to M1 macrophages [13]. Others propose that, in specific contexts, MAGs might support mitochondrial β -oxidation and reduce pro-inflammatory signaling. Reconciling these disparate observations requires a systematic interrogation of MAG–macrophage interactions at the molecular level. In particular, the precise mechanisms by which MAGs shape macrophage polarization and function especially in obese adipose tissue remain insufficiently understood. Some investigations suggest that elevated MAG levels could tip macrophages toward a more oxidative metabolism, potentially mitigating certain pro-inflammatory effects [7]. Conversely, other work implies that excessive MAGs may heighten inflammatory pathways, contingent on the microenvironment and the fatty acid composition of the monoglycerides [8]. Elucidating these nuances is crucial for developing interventions that can fine-tune macrophage function in obesity and metabolic disease.

Figure 3 (below) shows a conceptual model of how MAGs may enter macrophages, undergo enzymatic processing, and either supply fatty acids for oxidation or trigger receptor-mediated signaling pathways.

This figure encapsulates the dual nature of MAGs energy vs. inflammation and underscores their potential to modulate macrophage polarization.

What are the effects/impacts of MAG in adipose-tissue macrophage?

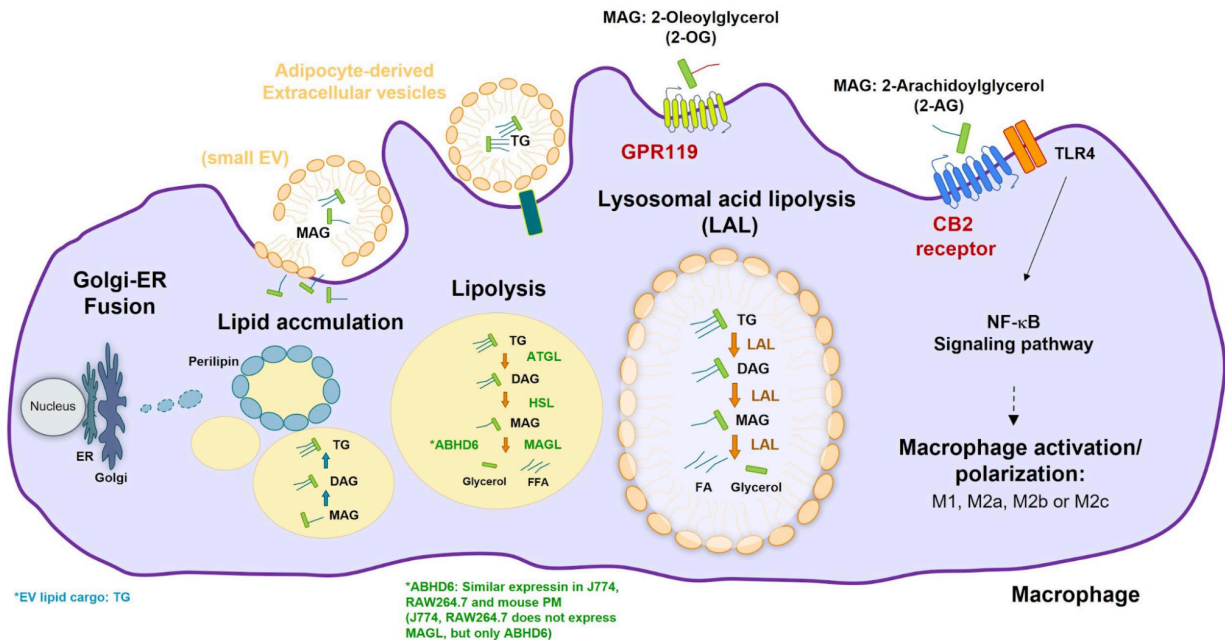


Figure 3: Schematic representation of how monoglycerides (MAGs) may influence adipose-tissue macrophages. Once taken up (left), MAGs such as 2-oleoyl glycerol (2-OG) or 2-arachidonoyl glycerol (2-AG) can be hydrolyzed to release free fatty acids (FFA) via lysosomal acid lipolysis (LAL), fueling lipid droplet formation or oxidative pathways. Meanwhile, certain MAGs engage receptors like GPR119 and CB2, activating NF-κB signaling and thus shifting macrophage polarization between M1 (pro-inflammatory) and M2 (anti-inflammatory) states. Adipocytes can also release MAG-laden vesicles, further modulating macrophage function. Collectively, these processes highlight the dual roles of MAGs—both as energy substrates and signaling molecules—in shaping macrophage metabolism and inflammatory tone within the obese adipose microenvironment.

Challenges in Studying Monoglyceride Metabolism: Spatial Lipidomics and Multiomic Integration

A major hurdle in dissecting monoglyceride (MAG) metabolism stems from the complexity of lipid metabolic networks and the highly interactive nature of adipose tissue. Macrophages do not handle

lipids in isolation; they simultaneously influence and are influenced by adipocytes, endothelial cells, and other immune cells within the adipose microenvironment [9]. Moreover, lipids can serve as energetic substrates, structural components, or signaling messengers, necessitating a multifaceted analytical approach to capture these roles [10,11]. Traditionally, researchers have employed single-omic strategies:

1. Transcriptomics: Identifies differentially expressed genes under various treatments, revealing which metabolic or inflammatory pathways might be upregulated or downregulated.
2. Bulk Lipidomics: Measures changes in the abundance of lipid classes—phospholipids, free fatty acids, triglycerides, and monoglycerides—across entire cell or tissue extracts.

While each of these methods yields valuable insights, using either one in isolation presents limitations. For instance, transcriptomic data can highlight candidate genes or enzymes (such as lipases or acyltransferases) responsible for processing MAGs, but offers no direct confirmation that these lipids are actually altered in the cell [9]. Meanwhile, bulk lipidomics may confirm that MAG levels rise or fall, yet it cannot definitively pinpoint which genes or pathways trigger those shifts nor indicate where in the cell (lipid droplets, mitochondria, ect.) the lipid accumulates [12]. Hence, a critical piece of the puzzle remains missing: the spatial context.

This is where spatial lipidomics emerges as a powerful tool. Techniques like matrix-assisted laser desorption/ionization imaging mass spectrometry (MALDI-IMS) allow researchers to visualize the distribution of lipids directly within tissue sections or cultured cells [13,14]. For adipose tissue macrophages, this spatial resolution is vital: the local microenvironment, be it crown-like structures around necrotic adipocytes or relatively healthy adipose regions, may profoundly affect how MAGs are metabolized or trigger downstream signaling [9]. If, for example, spatial lipidomics uncovers that a certain monoglyceride species accumulates in proximity to mitochondrial clusters, it could reinforce a transcriptomics-derived hypothesis suggesting enhanced β -oxidation. Conversely, localized buildup in lipid droplets might signal a storage or pro-inflammatory scenario. Furthermore, spatial data can reveal whether lipid alterations are uniform across the tissue or restricted to small, functionally significant domains, adding a nuanced layer to our understanding of macrophage-lipid interactions [10].

Bringing these data streams together under the umbrella of multi omic integration combining transcriptomics, lipidomics, and other large-scale datasets enables richer hypotheses and more robust conclusions [15,16]. If transcriptomics indicates upregulated enzymes catalyzing fatty acid β -oxidation, and bulk lipidomics confirms a concurrent rise in specific MAGs or their byproducts, it strongly supports

a direct metabolic pathway shift. Spatial lipidomics can then validate that these lipids localize in mitochondria-rich regions or near lipid droplets, offering a cause-and-effect chain from gene expression to actual metabolic changes [17]. Such comprehensive insights are crucial for adipose tissue macrophages, whose functional phenotypes may determine whether the tissue remains metabolically healthy or transitions into a chronically inflamed state [3,7,9]. Moreover, applying multi omics to monoglyceride metabolism specifically has the potential to reveal novel regulatory loops or signaling cascades that otherwise remain hidden in single-omic studies. Given the recognized importance of macrophage polarization in obesity, a deeper understanding of how monoglycerides modulate lipid storage and oxidation in these cells could lead to innovative strategies for reducing chronic inflammation or improving metabolic homeostasis in diseased adipose tissue [10,18].

In short, spatial lipidomics complements transcriptomics and bulk lipidomics by providing localized molecular signatures within the tissue or cell. Integrating all three data layers can illuminate how gene-level changes (seen in transcriptomics) correlate with shifts in lipid abundance (uncovered by bulk lipidomics) and their subcellular or tissue-specific localization (revealed by spatial lipidomics). Such a holistic, multi omic framework is crucial for discerning cause-and-effect relationships in macrophage lipid metabolism and may uncover entirely new avenues to fine-tune monoglyceride-driven processes in obesity.

Methods

Cell Culture and Treatments

Macrophage Line and Media

Murine J774A.1 macrophages (American Type Culture Collection, ATCC #TIB-67) were used for all experiments. Cells were maintained in high-glucose Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum (FBS) and 1% penicillin-streptomycin. Cultures were incubated at 37 °C in a 5% CO₂ humidified atmosphere. Cells were passaged at ~80% confluency, ensuring viability and consistent growth conditions.

Experimental Setup

When cultures reached approximately 80% confluency, the medium was gently removed and replaced with DMEM lacking FBS (to reduce background lipid interference). At the same time, cells were exposed to one of the following treatments for 24 hours:

1. Control (Vehicle): An equal volume of 95% ethanol corresponding to final solvent concentrations in treated groups.
2. Monoglyceride (MAG) Mixture: A 5 mM final concentration of a nonspecific MAG stock (purchased from Sigma Aldrich; cat. #1446000). This mixture contained a range of monoglyceride species as confirmed by preliminary LC–MS analysis (see Table 1).
3. Palmitic Acid (PA): A 5 mM final concentration, dissolved in 95% ethanol.
4. Oleic Acid (OA): A 5 mM final concentration, similarly dissolved in 95% ethanol.

The purpose of these treatments was to assess how different lipid substrates (specifically monoglycerides vs. free fatty acids) influence macrophage lipid metabolism and phenotype. After the 24-hour incubation, cells were either harvested for confocal imaging and subsequent lipid extraction (sections 1.3–1.4) or processed directly for RNA extraction in a parallel workflow (described in the transcriptomics section, not shown here).

Bodipy Staining and Confocal Imaging

Following the 24-hour treatment period, culture wells were carefully washed with 1× PBS (3 × 5-minute washes) to remove residual medium and unattached cells. Macrophages were then fixed using 4% paraformaldehyde for 20 minutes at room temperature, followed by three additional PBS washes.

Cells were stained with Bodipy (1 μ M in PBS) for 25 minutes at room temperature to visualize intracellular lipid droplets. After staining, each culture chamber was gently removed, and fluorescence-safe mounting media was placed onto the slides before adding coverslips. Slides were left to dry overnight, protected from light.

Confocal imaging was performed on a Leica SP8 confocal system under a 63× oil-immersion objective. Excitation and emission settings were optimized for Bodipy fluorescence (FITC channel), and Z-stacks were collected where relevant to visualize lipid droplet distribution in three dimensions. Representative images for each treatment group were captured and used to assess qualitative differences in lipid droplet content and cellular morphology.

Cell Sample Preparation for Lipid Extraction

Immediately after imaging (or in parallel with imaging samples), additional wells were dedicated to lipidomics. Media were removed, and cells were washed with 1× PBS (3 × 5 minutes). A volume of 300

μL methanol was added to each well or chamber, and the bottom of the plate was thoroughly scraped to dislodge macrophages. The resulting cell suspension was transferred to microcentrifuge tubes.

From each sample, 100 μL of the suspension underwent lipid extraction via a modified Bligh-Dyer method:

1. Initial Extraction: One volume (100 μL) of cell suspension was combined with 4 volumes (400 μL) of a 3:1 ratio of chloroform:methanol (adjusted from the classical Bligh-Dyer ratio to accommodate cell density).
2. Phase Separation: 0.8 volumes of water were added, and tubes were vortexed vigorously for 30 seconds. Samples were then centrifuged at $1,000 \times g$ for 5 minutes, forming an upper aqueous phase and a lower organic phase.
3. Organic Phase Collection: The lower organic phase, containing lipids, was carefully pipetted into a clean tube.
4. Dry Down: The collected lipid extract was dried under a gentle stream of nitrogen gas or in a vacuum concentrator.

Dried lipid films were stored at -80°C until further LC–MS analysis.

Instrumental Lipidomics Analysis

Sample Reconstitution

Immediately prior to LC–MS analysis, the dried lipid extracts were resuspended in 10 μL of a 4:3:1 ratio of isopropanol:acetonitrile:water, ensuring thorough vortexing for ~1 minute. This solvent composition was selected to maintain broad lipid solubility while minimizing artifactual adduct formation.

LC–MS Instrumentation

All LC–MS measurements were performed on an Agilent G6230 BA UPLC–TOF–MS system with electrospray ionization (ESI) (Agilent Technologies, Santa Clara, CA). The ultra-performance liquid chromatography used a Zorbax Eclipse Plus C18 column (4.6 mm $\times$ 50 mm, 3.5 μm bead size) maintained at 30°C . The mobile phase consisted of two solvents:

- Phase A: A 5:1:4 ratio of isopropanol:methanol:water containing 5 mM ammonium acetate and 0.1% acetic acid.
- Phase B: A 99:1 ratio of isopropanol:water, also with 5 mM ammonium acetate and 0.1% acetic acid.

An initial gradient of 95% A / 5% B was held for 1 minute, followed by a gradual increase of B to 95% over 10 minutes, and subsequently returned to starting conditions. The total run time, including column re-equilibration, was 15 minutes per sample. Flow rate was 0.3 mL/min.

Electrospray Ionization Settings

Electrospray ionization (ESI) was performed in either positive or negative mode depending on the class of lipids of interest:

- Ion Source Temperature: 300 °C
- Nitrogen Gas Flow: 13 L/min
- Sheath Gas: N₂ at 12 L/min
- Capillary Voltage: +3500 V (positive mode) or –3500 V (negative mode)
- Fragmentor Voltage: 120 V

Tuning was optimized to detect a broad range of lipid classes, including free fatty acids, phospholipids, and monoglycerides.

Cell Culture and Treatments

1M Arachidonic acid, and 2-AG stock solutions were created in the lab. Stock was diluted to a concentration of 5µM for cell treatment.

Cell sample preparation for metabolomics analysis

After culture and 24 hr treatment in chamber slides, media was removed and macrophages were washed with PBS for 5 minutes x 3. PBS was then removed and 100µl of methanol was added to each chamber. The bottom of the plate was scraped completely and the cell suspension was collected. Lipid extractions were performed in 1.5 mL microcentrifuge tubes using polypropylene positive displacement pipettes. Ten microliters of sample were mixed with 90 µL of extraction solvent, consisting of butanol:methanol (1:1) with 10 mM ammonium formate.

Each sample was then vortexed for 5 seconds and subsequently bath-sonicated for 1 hour, with the temperature maintained at 21 to 25 °C. Samples were then centrifuged (13,000 xg, 10 minutes, 20 °C), and the supernatant transferred into 1.5 mL glass sample vials (5190-9062) with 200 µL glass inserts (5183-2085). Samples were capped using PTFE/S caps (5185-5820).

Instrumental metabolomics analysis

All were run in an Agilent LCMS-QTOF 6230. The mobile phase of liquid chromatography was a binary phase consisting of A phase and B phase. A phase consisted of 10 mM ammonium formate, 5 µM Agilent deactivator additive (p/n 5191-3940) in 5:3:2 water: acetonitrile: 2-propanol. B phase consisted of 10 mM ammonium formate in 1:9:90 water: acetonitrile: 2-propanol. The column used for separation was a Zorbax Eclipse Plus C18, 2.1x100mm, 1.8µm (Agilent). Dual Electrospray ionization temperatures were 200°C, with drying gas flow rate of 10 L/min of nitrogen and nebulizer pressure at 50 psi. Injection volume was 5µL. Data collected in positive mode.

Transcriptomic Data Analysis

Library preparation and RNA-sequencing were performed by an external core facility (XYZ Core), generating paired-end reads on an Illumina platform. Raw sequencing data were quality-checked with FastQC (v0.11.9) and trimmed for adapters using Trimmomatic (v0.39). Clean reads were aligned to the mouse reference genome (GRCm38) with STAR (v2.7.3a), and gene-level counts were obtained using featureCounts (subread v1.6.3). All downstream statistical analyses and differential expression (DE) calculations were conducted in R (v4.x) using DESeq2 (v1.x) for normalization, variance shrinkage, and detection of DE genes. Post-DE filtering (ex. fold change cutoff, p-adjusted < 0.05) yielded the top gene candidates, which were subsequently uploaded to MetaboAnalyst (v5.0) for gene-annotation and functional enrichment analyses. This integration enabled a comprehensive overview of macrophage gene-expression changes correlating with lipid-modulated phenotypes.

Data Preprocessing and Feature Identification

Raw data were acquired via Agilent MassHunter software, and Mass Profiler Professional 10.8 was used for downstream processing. Spectral features were automatically and manually compared to an internal reference library of known lipids (by molecular weight and retention time). Agilent's

Qualitative Analysis workflows further aided in identifying potential monoglycerides (ex. MG(18:1), MG(16:0), MG(0:18:2)) by comparing exact m/z values with expected formula weights.

All features were then filtered (removing background signals, isotopic duplicates) and aligned across samples. Principal Component Analysis (PCA) was performed to visualize clustering of conditions (Control, MAG, PA, OA). Lipid species significantly altered were identified via ANOVA with Tukey's post hoc tests, focusing on changes that correlated with transcriptomic findings. Figure 1 was generated by using Paint S on MAC.

Literature Review

Macrophage Heterogeneity and Inflammatory Signaling in Adipose Tissue

Macrophages in adipose tissue exhibit a spectrum of activation states, extending beyond a simple M1/M2 dichotomy [1]. Recent transcriptomic analyses suggest resident vs. recruited macrophages may coexist and shift in proportion as obesity progresses [2]. Local factors such as adipocyte-derived free fatty acids, chemokines, and adipokines shape macrophage phenotype, but systemic influences (ex. hormones, circulating lipoproteins) can also modulate polarization [3]. In healthy, lean adipose tissue, macrophages typically display an M2-like (anti-inflammatory) profile, aiding tissue homeostasis and debris clearance [1]. However, as obesity develops, adipocytes enlarge and secrete signals that push macrophages toward an M1-like (pro-inflammatory) phenotype, characterized by elevated TNF- α , IL-6, and other cytokines that disrupt insulin signaling [2,3].

One hallmark of obesity-induced adipose inflammation is the formation of crown-like structures (CLSs) around necrotic or hypertrophic adipocytes, in which macrophages display heightened levels of pro-inflammatory mediators (ex. TNF- α , IL-1 β , IL-6) [4]. These cytokines reinforce insulin resistance, fueling a self-perpetuating cycle of local and systemic metabolic dysfunction [5]. Meanwhile, certain macrophage subpopulations in milder or early-stage obesity retain anti-inflammatory (M2-biased) features, secreting IL-10 or other factors that curb damage [3]. Understanding how macrophages transition among these functional states and discovering molecular signals that tip the balance remains a key challenge in managing obesity-associated inflammation and metabolic disease.

Monoglycerides (MAGs) as Regulatory Lipids

Beyond Fatty Acids and Triglycerides

Historically, research on macrophage lipid handling has focused on long-chain fatty acids, triglycerides, and cholesterol esters. However, monoglycerides (MAGs) have emerged as equally important regulatory lipids in the context of metabolic disease [4]. MAGs arise from partial hydrolysis of triglycerides or phospholipids and can act as:

- **Energy Substrates:** Under certain conditions, MAGs release free fatty acids, potentially channeling macrophages toward β -oxidation and dampening inflammatory responses [5].
- **Signaling Molecules:** Certain MAGs, such as 2-arachidonoylglycerol (2-AG), bind endocannabinoid receptors (CB2, GPR119), modulating lipid metabolism and immune pathways [5].

Divergent Effects in Obesity

In obese adipose tissue, excess MAGs may reinforce pro-inflammatory cues if combined with local stress signals, pushing macrophages toward an M1-like profile [4,6]. Conversely, MAGs can in some contexts promote oxidative metabolism, reducing inflammatory pathways by directing macrophages to utilize fatty acids for ATP production [1,9]. Such contradictions highlight the context dependence of MAG action, governed by the specific fatty acid chain, local cytokine milieu, and subcellular location.

Clinical and Experimental Perspectives

Pharmacological monoglyceride lipase inhibition has shown promise in mouse models of diet-induced obesity, improving insulin sensitivity and attenuating adipose inflammation [5]. This underscores the notion that MAG turnover influences macrophage-driven tissue dysfunction. Moreover, partial glycerides (MAGs, DAGs) may not be mere intermediates but active regulators of kinases or transcription factors central to the inflammatory cascade [4]. Harnessing this dual role fueling or alleviating inflammation could open new therapeutic avenues for tackling obesity-related metabolic disorders.

Multi-Omic Approaches to Decipher Macrophage Biology

Transcriptomics + Lipidomics: Finding Correlations

Multi-omic strategies have proven instrumental in dissecting the dual roles of lipids in macrophages, especially in lipid-rich environments like obese adipose tissue [6]. Transcriptomics (RNA-seq) can pinpoint up- or downregulated genes under, for instance, elevated MAG conditions, while

bulk lipidomics quantifies the abundance of specific lipid species (phospholipids, sphingolipids, MAGs, DAGs) [7]. By correlating gene-expression patterns with shifts in lipid composition such as an upregulated lipase gene paired with a decrease in its substrate MAG one can infer potential enzyme-substrate relationships and hypothesize functional links.

Yet, even the union of transcriptomics and lipidomics may overlook spatial nuances in a heterogeneous tissue. Macrophages in crown-like structures vs. macrophages in less inflamed areas may exhibit drastically different lipid usage [6]. Hence, researchers increasingly turn to spatially resolved methods to verify whether lipids localize in mitochondria-rich subcellular zones (suggesting oxidation) or in droplet-rich regions (signaling possible storage or inflammation).

Spatial Lipidomics: Adding the “Where” Dimension

Spatial lipidomics via matrix-assisted laser desorption/ionization imaging mass spectrometry (MALDI-IMS) or desorption electrospray ionization mass spectrometry (DESI-IMS)—provides molecular maps of lipid distributions in tissue sections or cultured cells [10,11]. In adipose macrophages, this approach can confirm whether MAG pools accumulate near necrotic adipocytes (often pro-inflammatory) or within more oxidative niches. Paré et al. (2019) showed, in a pulmonary infection model, that alveolar macrophages formed distinct lipid “hotspots,” illustrating how local environments shape lipid metabolism [6]. Applied to obesity, spatial lipidomics can validate whether hypothesized gene-lipid interactions identified in transcriptomics + bulk lipidomics actually manifest in inflamed vs. healthier zones of adipose tissue.

Heterogeneity and Sampling

One persistent issue is the cellular heterogeneity of adipose tissue. Macrophages share space with adipocytes, endothelial cells, fibroblasts, and other cell types, complicating the isolation of pure macrophage signals [1]. Even within the macrophage population, subgroups can display distinct transcriptomic or lipidomic profiles, underscoring the need for methods like single-cell RNA-seq or flow sorting [9]. Analyzing the interplay between macrophage-derived transcripts and localized lipid species demands robust data-integration pipelines that can handle multi-omic and spatial data in tandem.

Clinical Relevance

While multi-omic studies in mice have shed light on potential metabolic targets (ex. monoglyceride lipase, certain transcription factors), translating these findings to humans remains

challenging [2]. Adipose depot distribution (visceral vs. subcutaneous), dietary factors, and genetic backgrounds may all alter how MAG metabolism plays out. Nevertheless, a better understanding of MAG-driven macrophage polarization could guide precision interventions for example, small-molecule inhibitors of MAG-processing enzymes or dietary strategies tailored to reduce harmful MAG accumulation.

Toward a Comprehensive Mechanistic Model

By combining transcriptomics, lipidomics, and spatial lipidomics, researchers can build a more cause-and-effect framework for how monoglycerides modulate macrophages in obese adipose depots [8]. For instance, an upregulated acyltransferase gene might correlate with localized MAG buildups in pro-inflammatory niches, while a deficiency might correspond to decreased adipose inflammation. This integrated roadmap may uncover novel regulatory loops or highlight synergy between existing anti-inflammatory and metabolic therapies.

Synthesis and Implications for the Present Study

In sum, macrophage polarization in obesity arises from a complex interplay of lipid mediators, local tissue signals, and genetic regulation [1–3]. Monoglycerides (MAGs) exemplify this dual role: in some contexts, they promote β -oxidation and reduce inflammation; in others, they heighten pro-inflammatory responses [4–6]. While transcriptomics plus bulk lipidomics can identify potential gene-lipid correlations, they often miss the microenvironmental context crucial in obese adipose tissue [6–8]. Spatial lipidomics fills this gap by mapping exactly where lipids accumulate be it in crown-like structures or near mitochondria and confirming if the predicted functional changes align with real, localized phenomena [10,11]. Building on these insights, the present work adopts a multi-omic integration approach transcriptomics, lipidomics, and spatial lipidomics to dissect how MAG metabolism shapes macrophage function in obese adipose depots. By linking gene-level regulation, overall lipid abundance, and subcellular distribution, this study aims to clarify whether MAGs tip macrophages toward beneficial oxidation or harmful inflammation, potentially revealing new therapeutic avenues for mitigating obesity-related metabolic dysfunction.

Results

Transcriptomic Profiling at 5 mM (High Dose)

To assess how monoglycerides (MAGs) and specific fatty acids (oleic acid, OA; palmitic acid, PA) affect macrophage gene expression, we performed an RNA-seq comparison at 5 mM concentrations. Figure 4 shows a heatmap of the top 25 differentially expressed genes (DEGs) relative to control. Red indicates upregulation, while blue indicates downregulation compared to macrophages in the absence of added fatty acids or MAGs. The columns represent the different treatment groups OA, PA, and MAG where each triplicate is labeled accordingly. Clustering along both rows and columns illustrates that OA- and PA-treated macrophages each induce distinct gene expression changes, whereas MAG treatments yield a separate cluster.

Top 25 Genes Differential Expression vs Control (5mM Sample)

Heatmap showing differential expression of 25 genes across 24 samples. The color scale ranges from -2.7053 (blue) to 2.7053 (red), with 0 being white. The genes listed on the y-axis are: Gm43682, mt-Rnr2, Lmbrd2, Vasn, Chrna2, Gm10010, Kcnk7, 4933424M12Rik, Strip2, Gm36527, Prickle2, Rnase6, Cd244, Lifr, Cx3cr1, Tgfb1, Dmpk, Zfyve28, St6gal1, Cpt1a, A930001C03Rik, Gm26407, Mir1957a, Gm15443, and Gp1ba. The samples on the x-axis are grouped by treatment: Oleic Acid (3 samples), Palmitic Acid (3 samples), and MAG (3 samples). Hierarchical clustering dendrograms are shown on the top and left.

Figure 4: Heatmap of the top 25 differentially expressed genes (DEGs) vs. control at 5 mM. Red indicates upregulation; blue indicates downregulation. Each column represents a replicate under a specified treatment (OA, PA, or MAG).

Gene Ranking and Key Hits

To further compare the magnitude of differential expression changes, we plotted the average differential expression of these top 25 genes (Figure 5). Genes on the far right such as Dmpk exhibit the largest overall fold changes, whereas those closer to the left or center show more moderate alterations. This ranking view helps quickly identify which DEGs are most impacted by these treatments across replicates. Notably, Cpt1a also ranks high, underscoring its potential importance in monoglyceride- and fatty acid-treated macrophages. Collectively, these data provide a snapshot of genes that respond most strongly to changes in lipid environment at 5 mM concentrations.

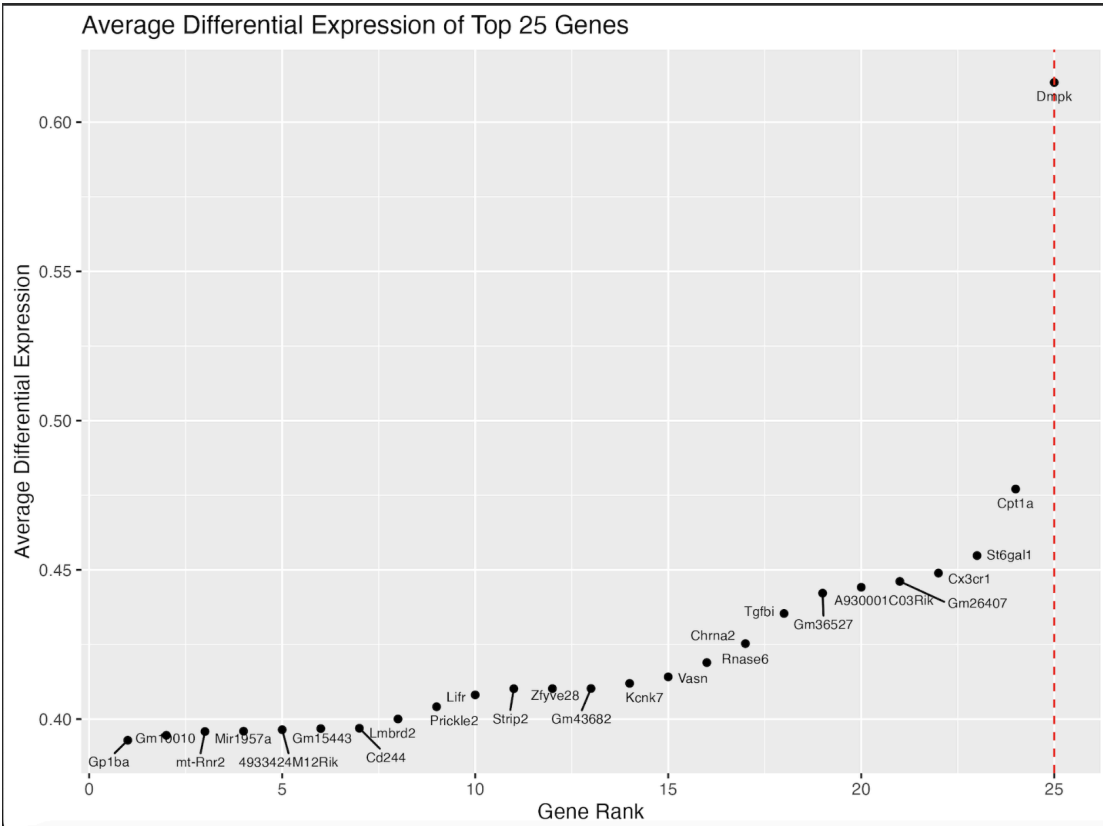


Figure 5: Rank plot showing the average differential expression for these 25 DEGs across all treatments. Genes at the far right (ex. Dmpk, Cpt1a) exhibit the highest overall fold changes.

Venn Diagram of 5 mM Conditions

We next explored the overlap of DEGs across three comparisons Control vs. MAG, Control vs. OA, and Control vs. PA using a Venn diagram (Figure 6). Each colored circle represents one comparison, and the counts indicate how many genes are uniquely or commonly altered. For instance, some genes appear in the center overlap, implying they are differentially expressed in all conditions (MAG, OA, and PA) relative to control. Others are exclusive to OA or PA, suggesting treatment-specific transcriptional signatures. Larger numbers in the OA circle (ex. 582) may reflect robust transcriptional shifts under oleic acid. By contrast, the MAG circle shows a different distribution, implying certain genes preferentially respond to monoglyceride exposure.

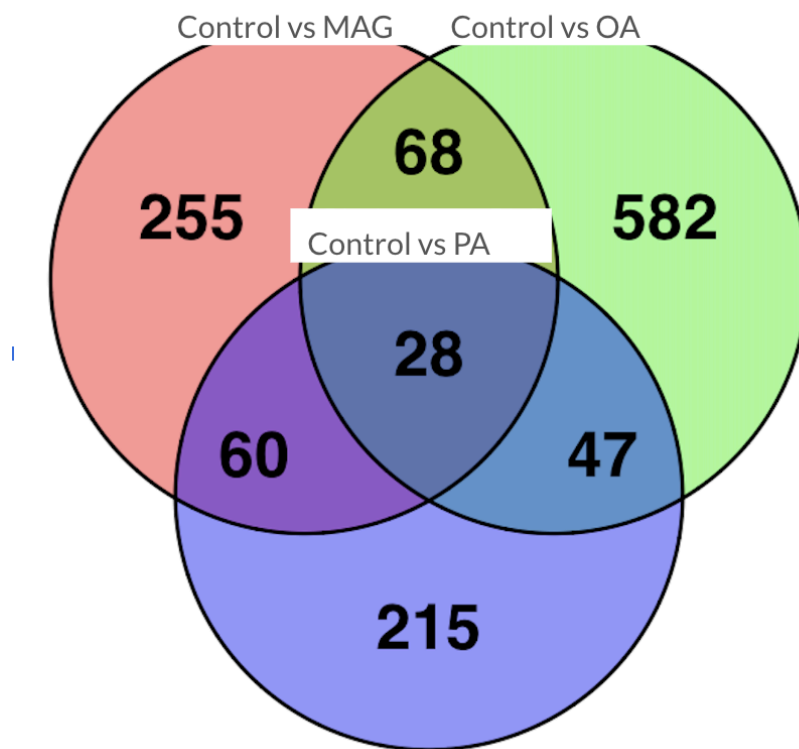


Figure 6: Venn diagram comparing Control vs. MAG, Control vs. OA, and Control vs. PA at 5 mM. Numbers within each region indicate how many genes are differentially expressed uniquely or in overlap among the three conditions.

Gene Lists for OA+MAG and PA+MAG Overlaps

To dissect the functional themes of genes that simultaneously change under OA+MAG or PA+MAG, we compiled lists (Figure 7) focusing on 5 mM overlaps. The OA+MAG overlap includes genes

involved in lipid metabolism, inflammation, and tissue remodeling, for example *Angptl4*, *Cpt1a*, and *Notch4* suggesting a dual role in immune regulation and metabolic adaptation. Meanwhile, the PA+MAG overlap reveals genes tied to immune signaling, transport mechanisms, and possibly cell survival (ex, *Acot10*, *Dusp13*, *Tnfrsf13c*). Notably, the presence of *Cpt1a* in the OA+MAG subset reinforces the notion that monoglycerides may enhance fatty acid oxidation pathways, though the ultimate functional outcome likely depends on additional environmental cues.

PA + MAG Overlap (5mM)		OA + MAG Overlap (5mM)	
Gene	Function	Gene	Function
<i>Acot10</i>	Fatty acid metabolism	<i>Angptl4</i>	Lipid metabolism, inflammation
<i>Bmx</i>	Cell signaling/survival	<i>Cpt1a</i>	Fatty acid oxidation
<i>Dusp13</i>	MAPK phosphatase	<i>Ido1</i>	Immune regulation
<i>Fosb</i>	Stress/immune transcription factor	<i>Il17ra</i>	Cytokine signaling
<i>P2rx6</i>	Purinergic receptor	<i>Plin2</i>	Lipid droplet formation
<i>Plppr3</i>	Phospholipid phosphatase	<i>Mmp8</i>	Matrix remodeling
<i>Tm6sf2</i>	Cholesterol metabolism	<i>Trpm1</i>	Ion transport/sensing
<i>Tnfrsf13c</i>	TNF receptor superfamily	<i>Notch4</i>	Notch signaling
<i>Slco4a1</i>	Solute carrier	<i>Rnf144b</i>	Protein degradation

Figure 7: Genes Overlapping in OA+MAG and PA+MAG at 5 mM. Each table highlights genes co-regulated by monoglycerides (MAGs) and either oleic acid (OA) or palmitic acid (PA). Functional annotations indicate roles in lipid metabolism, immune signaling, and tissue remodeling. These “Function” annotations come from Public Gene Databases such as Ensembl or NCBI Gene.

Transcriptomic Profiling at 500 μ M (Low Dose)

Venn Diagram of 500 μ M Conditions

In contrast to the 5 mM treatments, we also examined macrophages exposed to 500 μ M concentrations of MAG, OA, or PA. Figure 8 displays a Venn diagram for the Control vs. MAG, Control vs. OA, and Control vs. PA comparisons at this lower concentration. As before, the diagram partitions DEGs into regions of unique and shared differentially expressed genes. Notably, the circles for OA and PA each contain substantial unique segments (ex. 195 for Control vs. MAG in one condition, 473 for Control vs. OA in another), indicating that even at 500 μ M, these fatty acids elicit distinct transcriptional profiles. The smaller overlap in the center suggests fewer genes are altered by all three treatments simultaneously at this dose, implying the 5 mM treatments may drive more extensive or uniform changes than 500 μ M.

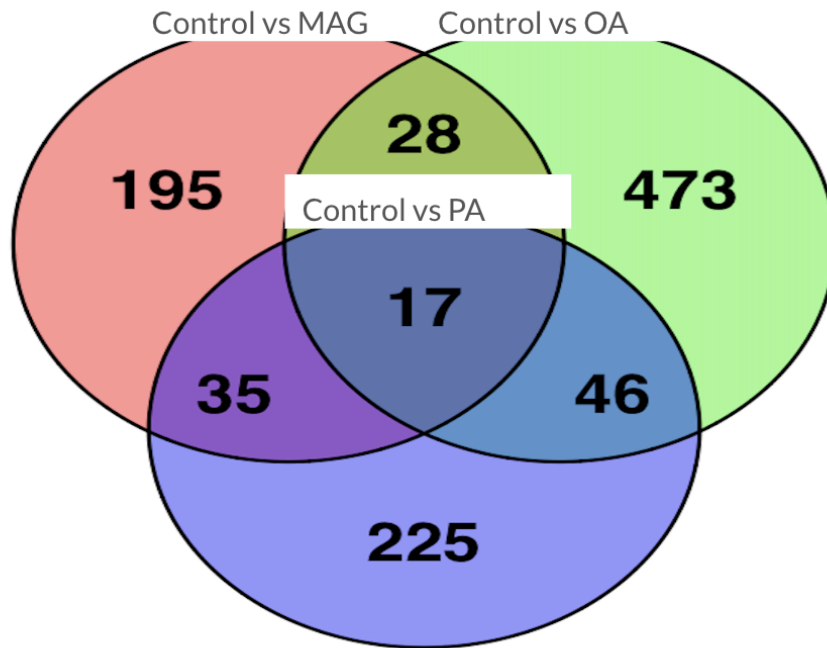


Figure 8: Venn diagram comparing Control vs. MAG, Control vs. OA, and Control vs. PA at 500 μ M. Numbers within each region indicate how many genes are differentially expressed uniquely or in overlap among the three conditions.

Gene Lists for OA+MAG and PA+MAG Overlaps

The gene list for the overlaps at 500 μ M is shown below. Under OA+MAG, the genes (ex. Cxcl11, Alox12e, Hdac11) predominantly relate to inflammation, metabolism, or tissue remodeling. In the PA+MAG overlap, transcripts such as Ccr6, Fut1, Gprc5b, and Lgr6 are implicated in immune responses, cell signaling, and potential stem cell regulation. Taken together, these data hint at a more moderate but still physiologically relevant shift in macrophage gene expression at 500 μ M, possibly skewing cells toward remodeling or inflammatory pathways depending on the specific fatty acid or monoglyceride.

OA + MAG Overlap (500µM)		PA + MAG Overlap (500µM)	
Gene	Function	Gene	Function
Cxcl11	Chemokine/inflammatory signaling	Ccr6	Chemokine receptor (immune response)
Alox12e	Fatty acid/lipid metabolism	Fut1	Fucosyltransferase (glycosylation)
Hdac11	Histone deacetylase, epigenetic regulation	Gprc5b	Cell signaling receptor, metabolic regulation
Acer2	Ceramidase/lipid metabolism	Wnt5b	Wnt signaling, tissue remodeling
Aspn	Extracellular matrix (ECM) remodeling	Sema3g	Semaphorin signaling, axon guidance/cell migration
Adams15	Matrix protease, ECM remodeling	Serpinb10	Serine protease inhibitor (inflammation)
		Rxfp2	Relaxin receptor, reproductive/vascular function
		Lgr6	Stem cell receptor, tissue remodeling

Figure 9: Genes Overlapping in OA+MAG and PA+MAG at 500 uM. Each table highlights genes co-regulated by monoglycerides (MAGs) and either oleic acid (OA) or palmitic acid (PA). Functional annotations indicate roles in lipid metabolism, immune signaling, and tissue remodeling. These Function annotations come from Public Gene Databases such as Ensembl or NCBI Gene.

Summary of Transcriptomic Findings

Overall, both 5 mM and 500 µM treatments with monoglycerides, oleic acid, and palmitic acid induce distinct gene expression patterns in macrophages, with some overlap suggesting common regulatory nodes particularly those tied to immune response and lipid handling. The top 25 differentially expressed genes at 5 mM highlight potential key players like Dmpk and Cpt1a, while the Venn diagrams reveal unique and shared pathways modulated by OA or PA in the presence of MAGs. Ongoing analyses, including functional enrichment and correlation with lipidomics data, will further clarify how these DEGs might drive macrophage phenotypes in the context of lipid-rich environments.

Spatial Lipidomics Results

Top Five Monoglycerides (MGs) Detected in Obese Adipose Tissue via MALDI Imaging Mass Spectrometry

To understand how specific monoglyceride (MG) species distribute within obese adipose tissue, we performed MALDI imaging mass spectrometry (IMS) on tissue sections. This approach enables high-resolution spatial mapping of individual MGs, providing insight into which regions of the tissue (ex. crown-like structures, vascular areas) may accumulate these glycerolipids. Identifying “hotspots” for each MG can guide subsequent analyses linking lipid presence to macrophage infiltration or local inflammatory processes.

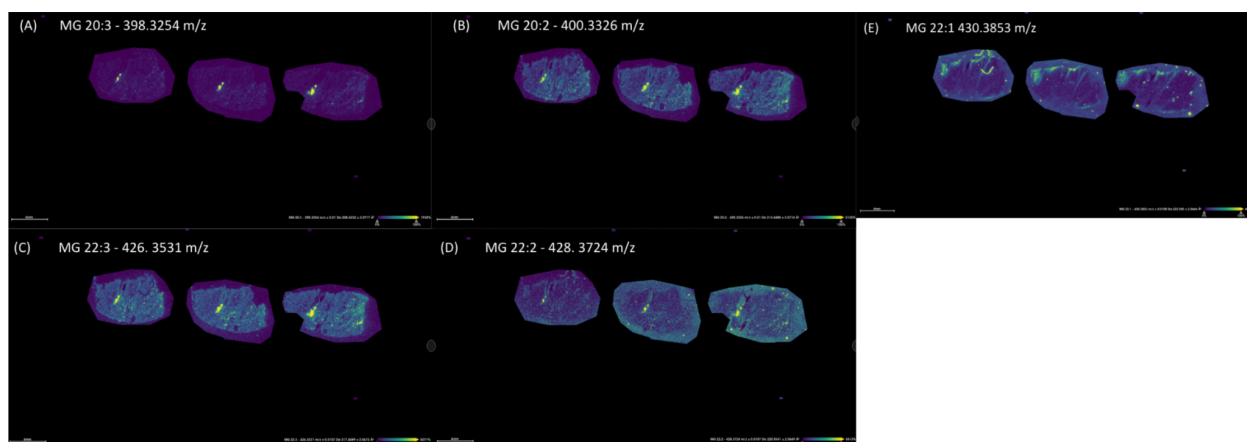


Figure 10: Spatial Ion Images of the top five most abundant monoglyceride (MG) species identified in obese adipose tissue. Panels (A)–(E) correspond to MG(20:3) (m/z 398.3254), MG(20:2) (m/z 400.3326), MG(22:3) (m/z 426.3531), MG(22:2) (m/z 428.3724), and MG(22:1) (m/z 430.3853), respectively. Warmer colors indicate higher local signal intensity, auto-scaled for each MG to highlight “hotspots” that may coincide with inflamed regions. The three tissue sections in each panel represent biological replicates or distinct areas of the same tissue block.

Top Four Triglycerides (TGs) Detected in Obese Adipose Tissue via MALDI Imaging Mass Spectrometry

To complement the monoglyceride (MG) analysis, we also profiled the spatial distribution of triglycerides (TGs) the predominant storage form of fatty acids in adipose tissue. By applying MALDI imaging mass spectrometry (IMS), we can pinpoint regions of intense TG accumulation within obese adipose samples, potentially unveiling lipid hotspots that coincide with immune cell infiltration or necrotic adipocytes. Visualizing the most abundant TG species offers insights into how stored fats may impact local inflammation and metabolic regulation in adipose depots characterized by obesity.

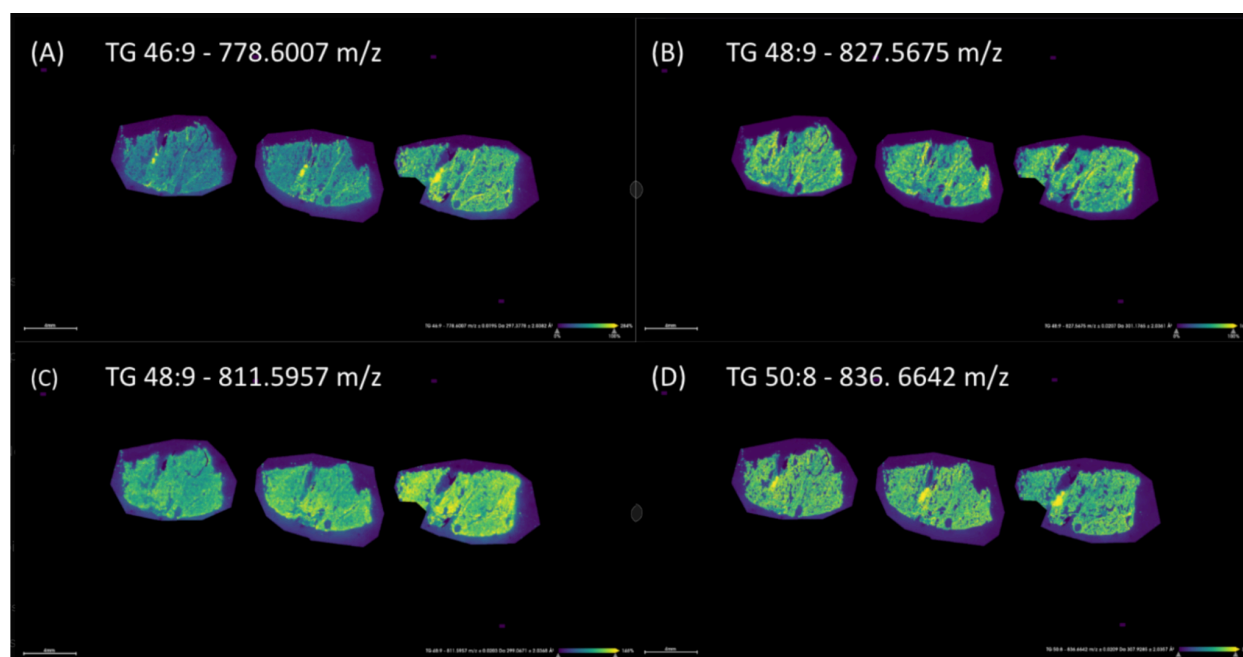


Figure 11: Spatial Ion Images of the top four most abundant triglyceride (TG) species identified in obese adipose tissue. Panels (A)–(D) correspond to TG(46:9) (m/z 778.6007), TG(48:9) (m/z 827.5675), TG(48:9) (m/z 811.5957), and TG(50:8) (m/z 836.6642), respectively. Warmer colors in each panel denote higher local signal intensity (auto-scaled to highlight distinct hotspots across the tissue). The three tissue sections per panel represent biological replicates or different regions from the same adipose depot.

Lipidomics Results

PCA of Global Lipid Profiles

To assess the overall lipidomic landscape in response to arachidonic acid (AA) and 2-arachidonoylglycerol (2-AG), a Principal Component Analysis (PCA) on the full set of lipid features was performed. By reducing the dimensionality of the data, PCA helps visualize broad metabolic shifts across treatments and identifies whether samples cluster distinctly based on treatment type. Notably, this approach reveals whether AA- or 2-AG-treated samples diverge from vehicle controls, providing insight into how each fatty-acid-derived compound influences the global lipid profile.

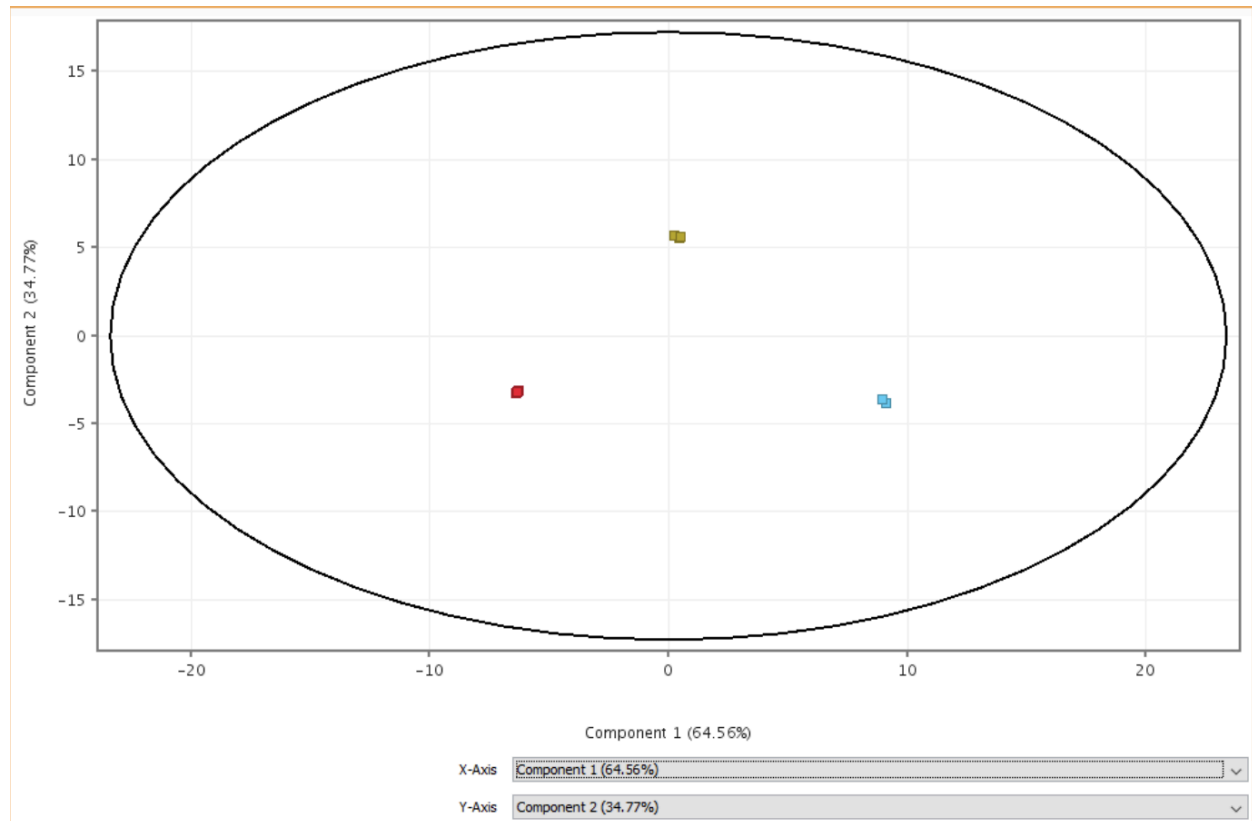


Figure 12: Principal Component Analysis (PCA) of Global Lipid Profiles

PCA score plot showing the distribution of samples treated with arachidonic acid (red), 2-arachidonoylglycerol (2-AG; blue), and vehicle (yellow) along the first two principal components. Component 1 (64.56%) and Component 2 (34.77%) together explain the majority of the variance in the data, illustrating how each treatment group clusters based on its overall lipidomic signature. Samples projecting closer in PCA space exhibit more similar global lipid compositions, whereas greater distance indicates larger metabolic differences. The ellipse denotes a 95% confidence region, aiding visualization of group clustering. This analysis provides a broad overview of the lipidomic shifts induced by arachidonic acid vs. 2-AG, highlighting any potential separation from the vehicle control.

Heatmaps of Differential Lipid Features

To further dissect how arachidonic acid (AA) and 2-arachidonoylglycerol (2-AG) treatments alter macrophage lipid profiles, we generated hierarchical clustering heatmaps focusing on the most variable or biologically relevant lipids. By comparing these heatmaps, we can visualize how each treatment group (AA, 2-AG, or vehicle) clusters based on up- or downregulated lipid intensities. This allows us to pinpoint which lipid classes or individual species diverge most under each condition and whether 2-AG-treated samples mirror vehicle more closely than AA-treated samples. Ultimately, these heatmaps can guide

deeper investigations into specific pathways (ex. eicosanoid, endocannabinoid) implicated in obesity-related inflammation and metabolic shifts.

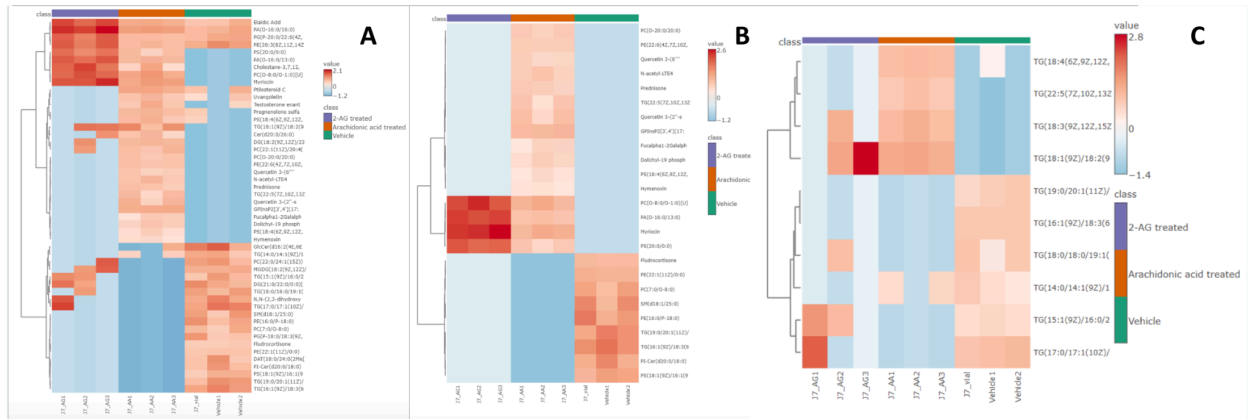


Figure 13: Hierarchical Clustering Heatmaps comparing 2-AG–treated (purple), arachidonic acid–treated (orange), and vehicle (green) samples. Each panel presents a distinct subset of the most variable or relevant lipids, with warmer (red) cells indicating higher relative abundance and cooler (blue) cells indicating lower abundance:

- **(A) Top 25 Lipids:** A focused set of 25 lipid species ranked by variance or statistical significance, showcasing clear separations among AA, 2-AG, and vehicle groups. Note how certain phospholipids or fatty acids appear enriched in AA–treated samples.
- **(B) Top 50 Lipids:** An expanded view of 50 highly variable or significantly changing analytes. Here, 2-AG–treated replicates sometimes cluster nearer to the vehicle, suggesting partial overlap in lipid profiles compared to the more distinct AA–treated group.
- **(C)** Contains an additional subset (ex. another 25/50/100 lipids) of particular interest; these may capture unique triglycerides, diglycerides, or a specialized set of bioactive lipids not covered in Panels (A) or (B). Column dendrograms depict sample clustering, while row dendrograms reflect how similarly lipids co-vary across treatments.

Columns represent individual samples, and rows represent specific lipid analytes. Branch lengths in the dendrograms reflect how similarly (or differently) samples and lipids group, offering a global overview of how AA vs. 2-AG treatments shape the macrophage lipidome relative to vehicle controls.

Changes in Specific Lipid Classes

To further pinpoint which lipid classes underlie the global shifts observed in the PCA and heatmap analyses, we measured the abundance of individual lipid classes in samples treated with arachidonic acid (AA) or 2-arachidonoylglycerol (2-AG), comparing them against vehicle controls. This targeted approach allows us to identify whether particular classes such as monoglycerides (MGs), triglycerides (TGs), ceramides, or phospholipids show pronounced up- or downregulation. By examining these bar charts, we can deduce which pathways or enzymatic processes may be differentially activated by AA vs. 2-AG, providing mechanistic clues about how endocannabinoid- and eicosanoid-related lipids modulate metabolic and inflammatory responses.

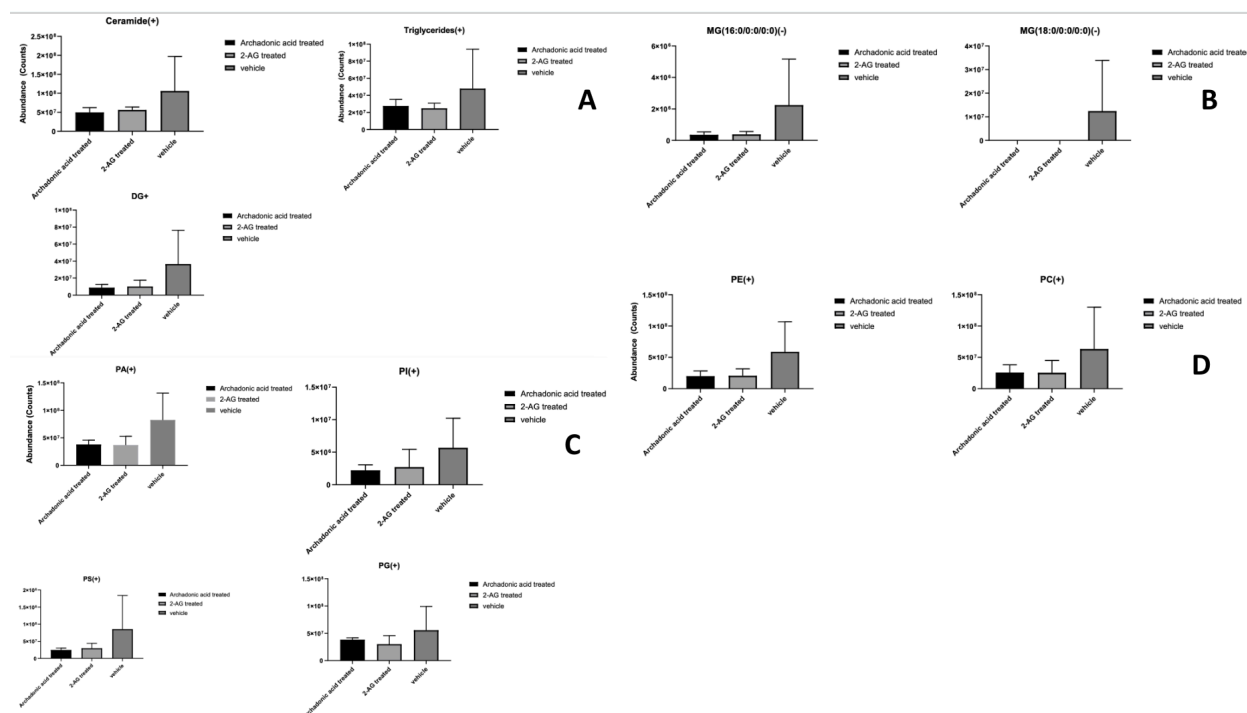


Figure 14: Abundance of Selected Lipid Classes in Arachidonic Acid, 2-AG, and Vehicle-Treated Samples.

Each bar chart compares mean lipid intensities $\pm$ SEM (or SD) for *n* replicates (ex. three biological replicates per treatment). The x-axes list treatment groups, while the y-axes reflect absolute or relative ion intensities (counts) measured by LC–MS.

- **(A) Ceramide(+)** and **Triglycerides(+)**: Illustrates whether ceramide, a sphingolipid often linked to insulin resistance, and TGs, the main storage form of fatty acids, are elevated or reduced under each condition.

- **(B) MG(16:0/0:0/0:0) and MG(18:0/0:0/0:0):** Focuses on specific monoglycerides containing palmitic (16:0) or stearic (18:0) fatty acids, indicating potential differences in partial glyceride metabolism.
- **(C) DG(+), PA(+), PI(+), PS(+), PG(+):** Captures diglycerides (DG) and several phospholipid classes phosphatidic acid (PA), phosphatidylinositol (PI), phosphatidylserine (PS), and phosphatidylglycerol (PG) to see how they respond to AA or 2-AG treatment.
- **(D) PE(+) and PC(+):** Shows phosphatidylethanolamine (PE) and phosphatidylcholine (PC), two major membrane phospholipids that may differ among treatments and influence membrane properties or lipid signaling.

In all sub-panels, black bars represent AA-treated samples, gray bars are vehicles, and light-gray (or white) bars indicate 2-AG–treated samples (or vice versa, per your legend). Any observed increase or decrease in abundance suggests that a particular lipid class could be selectively modulated by these treatments, highlighting potential pathways for further functional or mechanistic exploration.

Discussion

Gene Expression Shifts in Macrophages Exposed to Monoglycerides and Fatty Acids

The transcriptomic data presented offer a multifaceted view of how monoglycerides (MAGs) and specific fatty acids oleic acid (OA) and palmitic acid (PA) alter macrophage gene expression under both 5 mM and 500 μ M treatment conditions. By comparing these exposure levels, we can begin to discern dose-dependent patterns in immune responses, lipid metabolism, and tissue remodeling pathways, while also identifying certain overlaps in regulated genes that suggest shared mechanisms. These findings are crucial in the context of obesity-linked inflammation [2,3], where adipose macrophages confront diverse lipid environments. Understanding how MAGs compare to free fatty acids (FFAs) helps unravel whether these lipids function as mere metabolic substrates or actively drive immune signaling [4,5].

High Dose (5 mM) Lipid Treatments Elicit Distinct Macrophage Transcriptional Profiles

At 5 mM, we observed robust transcriptional shifts across all treatments. The heatmap of the top 25 differentially expressed genes (DEGs) revealed distinct clustering by condition, with OA and PA eliciting unique expression signatures, whereas MAG-treated macrophages formed a separate cluster.

From a methodological standpoint, this clustering underscores that each lipid microenvironment MAG, OA, or PA can produce a characteristic molecular fingerprint in macrophages [3], likely reflecting different receptor usage (ex. GPCRs vs. TLRs), intracellular signaling cascades, and metabolic outcomes [1].

Notably, the two genes *Dmpk* and *Cpt1a* stood out for their pronounced upregulation in certain conditions:

- **Dmpk:** Exhibited the highest overall fold change in the rank plot, suggesting potential involvement in an extreme metabolic or stress-adaptive response. Although *Dmpk* is often associated with muscle function, its strong transcriptional response in macrophages may reflect an underappreciated role in cytoskeletal or signaling pathways modulated by the lipid environment. One possibility is that *Dmpk* helps reorganize the cytoskeleton or modifies certain kinase-dependent pathways in lipid-loaded macrophages, shaping their morphology or migratory capacity in obese adipose tissue [3].
- **Cpt1a:** Known for its role in fatty acid β -oxidation, *Cpt1a* was also highly upregulated, implying that increased lipid availability particularly under MAG or OA conditions may drive macrophages toward enhanced fatty acid utilization or oxidation. Such shifts could reshape the energetic and inflammatory landscape within adipose tissue with obesity, where macrophage polarization is often dictated by metabolic cues. This result aligns with growing evidence that macrophage function is intimately linked to cellular metabolism [1]. By boosting β -oxidation pathways, macrophages could either become more “anti-inflammatory” (in an M2-like sense) or experience a unique metabolic phenotype in the presence of these lipids.

The Venn diagram at 5 mM indicated that while each treatment (MAG, OA, PA) shares some DEGs, many genes remained unique to a specific condition. For instance, the OA circle displayed a notably large subset (ex. 582 genes) not shared with PA or MAG, implying that oleic acid may trigger a pronounced transcriptional program emphasizing lipid handling and possibly pro- or anti-inflammatory pathways, depending on the microenvironment [2,3]. Meanwhile, MAG-specific genes suggest that monoglycerides can exert distinct signals potentially through specialized lipases or GPCR pathways not activated by free fatty acids alone [4,5]. This highlights the specificity of macrophage responses: OA might favor certain inflammatory or metabolic genes less triggered by PA, while MAGs might tap into

alternative receptor systems or enzyme cascades. Such distinctions are essential for unraveling how real-world adipose tissue, which contains both FFAs and partial glycerides, orchestrates macrophage polarization [1,3].

Synergistic Effects of Monoglycerides and Fatty Acids at 5 mM

To delve deeper into how monoglycerides and fatty acids might cooperate, we examined overlapping DEGs under OA+MAG and PA+MAG. Genes in the OA+MAG overlap (ex. *Angptl4*, *Cpt1a*, *Notch4*) appear tied to lipid metabolism, inflammation, and tissue remodeling, reflecting a dual push toward metabolic adaptation (via *Cpt1a*) and immune regulation (via *Angptl4*, *Notch4*). Meanwhile, PA+MAG overlaps involve genes such as *Acot10*, *Dusp13*, *Tnfrsf13c*, implicating immune signaling, phosphatase-mediated regulation, and possible cell survival pathways [4,5]. Collectively, these results hint that monoglycerides can synergize with either OA or PA to modulate inflammatory or remodeling processes more intricately than either fatty acid alone. This synergy is especially intriguing because it suggests that even within the same dosage, pairing MAG with OA or PA leads to partially diverging gene sets. For instance, *Angptl4* influences angiogenesis and lipid metabolism, whereas *Tnfrsf13c* (also known as B-cell maturation antigen receptor) can guide immune cell cross-talk. Such synergy could be pivotal in crown-like structures (CLSs) where lipid-laden macrophages accumulate in obese adipose depots [2,3].

Low Dose (500 μ M) Exposures Reveal Subtler Gene Expression Changes

Despite the generally diminished transcriptional response, the OA+MAG and PA+MAG overlaps at 500 μ M still highlight specific genes with meaningful functional implications. On the OA+MAG side, transcripts such as *Cxcl11*, a chemokine tied to inflammatory signaling and macrophage recruitment [5]; *Alox12e*, involved in fatty acid/lipid metabolism and potentially influencing local eicosanoid production [4]; and *Hdac11*, a histone deacetylase implicated in epigenetic regulation, collectively suggest possible remodeling of macrophage gene expression at the chromatin level. Meanwhile, in the PA+MAG overlap, genes like *Ccr6*, a chemokine receptor steering immune or inflammatory responses; *Fut1*, a fucosyltransferase that could modify cell surface glycosylation and impact receptor-ligand interactions; and *Lgr6*, a receptor linked to stem cell functions and tissue remodeling, illustrate additional mechanisms whereby these lipids may modulate macrophage behavior [2,8]. Even though the number of DEGs is lower at 500 μ M, it's notable that these genes remain functionally rich: chemokines, metabolic enzymes, and epigenetic modifiers can have outsized impacts on how macrophages adapt to lower lipid

challenges [9]. The presence of epigenetic elements like Hdac11 suggests that changes in histone acetylation status may prime macrophages for persistent phenotypic shifts, bridging short-term lipid uptake with long-term reprogramming [1,9].

Together, these gene sets propose a moderate yet still biologically relevant shift in macrophage gene expression at 500 μ M encompassing chemokine signaling, lipid processing, epigenetic regulation, and matrix remodeling [9]. Such processes may enable macrophages to adapt to a lower lipid challenge by skewing more subtly toward either a remodeling or inflammatory phenotype, depending on the fatty acid or monoglyceride present [4,5]. These subtler shifts are consistent with a dose-dependent model, in which macrophages sense and respond proportionately to the concentration of lipids in their environment. In vivo, where lipid concentrations can fluctuate, this dose sensitivity might determine the overall inflammatory or remodeling tone in adipose tissue [3,10].

Integrating Lipid-Driven Transcriptomic Shifts with Obesity-Related Inflammation

Overall the data imply that the lipid environment, whether dominated by monoglycerides or free fatty acids, strongly influences macrophage phenotypes, potentially steering them between pro- and anti-inflammatory states and shifting tissue remodeling programs [2,3,5]. The robust upregulation of Dmpk at 5 mM may reflect a stress-adaptive or signaling pathway rarely highlighted in macrophage studies, warranting further investigation. Moreover, partial gene overlaps in OA+MAG and PA+MAG conditions suggest that monoglycerides can synergize with each fatty acid to enact specialized remodeling or immune signaling circuits a synergy that becomes subtler yet still detectable at 500 μ M [4]. This synergy is particularly relevant to pathological states of obesity, where multiple lipid species co-exist in interstitial fluid, fostering complexities in macrophage activation [3]. If, for instance, monoglycerides potentiate or modulate the inflammatory impetus of FFAs, therapeutic interventions might aim to regulate MG formation or specific enzyme pathways (ex. MG lipases) to reduce maladaptive inflammation [5].

Taken together, these findings expand our view of how monoglycerides (MAGs) and fatty acids like oleic acid (OA) and palmitic acid (PA) shape the transcriptome of adipose macrophages. They also underscore the likelihood that macrophage polarization in obese adipose tissue is orchestrated by multiple lipid-derived cues, with both concentration and lipid type (MAG vs. FFA) dictating distinct pathways [4,8]. Further mechanistic assays coupled with comprehensive lipidomics analyses will be

essential for determining how these genes and pathways translate into functional macrophage phenotypes relevant to obesity-induced inflammation and insulin resistance [9,10]. Exploring whether these gene signatures persist in vivo perhaps by comparing crown-like structures in lean vs. obese adipose tissue could validate the clinical relevance of these in vitro observations [2,3]. Ultimately, unraveling how MAGs and FFAs jointly drive macrophage phenotypes holds promise for refining metabolic or immunotherapeutic strategies to combat obesity-related disorders.

Spatial Lipidomics: Mapping Monoglyceride and Triglyceride Hotspots in Obese Adipose Tissue

Utilizing MALDI imaging mass spectrometry (IMS) to identify and visualize the top five monoglyceride (MG) species—MG(20:3), MG(20:2), MG(22:3), MG(22:2), MG(22:1)—and the top four triglyceride (TG) species—TG(46:9), TG(48:9), TG(48:3), TG(50:8)—in obese adipose tissue. Spatial ion images highlight distinct “hotspots” for each MG/TG, often overlapping with macrophage-rich crown-like structures (CLSs) [20].

In these MGs and TGs, the glycerol backbone is esterified to specific fatty acids. For monoglycerides, only one fatty acid is attached, whereas triglycerides have three. Understanding which fatty acid moiety (ex. eicosatrienoic acid, docosatrienoic acid) is present can offer clues to whether a lipid might be pro- or anti-inflammatory in nature, given that different polyunsaturated fatty acid (PUFA) chains can yield distinct eicosanoid or docosanoid mediators [10]. For instance, MG(20:3) contains a 20-carbon chain with three double bonds (likely eicosatrienoic acid), while MG(22:3) and MG(22:2) involve 22-carbon chains potentially corresponding to docosatrienoic or docosadienoic acids, respectively. These localized distributions suggest that certain glycerolipids accumulate preferentially where immune cells congregate, an observation that may signal metabolic or inflammatory cross-talk at the tissue level [6,7]. By mapping out each lipid’s spatial intensity, we can infer whether particular MG/TG species are tied to necrotic adipocytes or supportive stromal areas, offering a more nuanced perspective than bulk lipid measurements alone.

Monoglycerides (MGs): Localized Modulators of Adipose Inflammation

In Figure 10, MG(20:3) and MG(22:2) exhibited especially prominent hotspots, while MG(20:2), MG(22:3), and MG(22:1) showed moderately higher intensities in perivascular regions. Notably,

MG(20:3) (A) and MG(22:2) (D) display pronounced focal enrichment in certain subregions, possibly indicative of heightened local monoacylglycerol metabolism or immune cell activity. MG(20:2) (B), MG(22:3) (C), and MG(22:1) (E) show moderately increased signal near perivascular zones, suggesting that each MG species may respond to or influence distinct microenvironmental cues in obese adipose depots. These results underscore the potential for monoglycerides to serve not only as metabolic intermediates but also as localized modulators of adipose inflammation [10].

Structurally, MG(20:3) suggests the fatty acid is eicosatrienoic acid, which may be an n-6 derivative like dihomono- γ -linolenic acid (DGLA; 20:3 n-6) or Mead acid (20:3 n-9) if essential fatty acids are deficient. DGLA can be metabolized into less pro-inflammatory eicosanoids compared to arachidonic acid (20:4), hinting at a possible anti-inflammatory or modulatory role. Meanwhile, MG(22:3) points to docosatrienoic acid, a 22-carbon chain with three double bonds though relatively underexplored, it may represent an elongated PUFA that influences local macrophage function. MG(22:2) (docosadienoic acid) and MG(22:1) (docosenoic acid) could arise from partial β -oxidation or elongation-desaturation pathways, with docosenoic acid (ex., erucic acid, 22:1 n-9) sometimes linked to membrane fluidity and potential mild pro- or anti-inflammatory signaling, depending on the isomer. Importantly, these MGs have only one acyl chain on glycerol, which can be quickly hydrolyzed or used for local signaling in ways that free fatty acids (FFAs) might not.

Depending on whether these are n-3, n-6, or n-9 series fatty acids, their downstream metabolites might exhibit anti-inflammatory (DGLA $\rightarrow$ PGE1) or pro-resolving properties although some chain variants can be neutral or even slightly pro-inflammatory. The heterogeneity in chain length and unsaturation suggests that certain MG species could serve as “danger signals” to macrophages, while others might buffer inflammation by fueling β -oxidation or by producing less harmful lipid mediators [10]. Such heterogeneity strongly suggests that monoglycerides are not merely transient intermediates but may exert local immunomodulatory effects particularly if they co-localize with macrophages. Confirming these spatial patterns with histological or immunofluorescence stains for macrophage markers (ex. CD68) could substantiate whether MG deposits consistently align with immune cell clusters [20].

Triglycerides (TGs): Species-Specific Distributions and Immune Interactions

In Figure 11, TG(46:9) showed strong intensity near CLS cores, whereas TG(48:9) appeared in two distinct m/z variants (B) 827.5675 and (C) 811.5957 suggesting isomeric forms or alternative adducts. TG(48:3) and TG(50:8) showed broader, diffuse patterns across the tissue. TG(46:9) (A) displays pronounced enrichment near putative crown-like structures, whereas TG(50:8) (D) exhibits a more diffuse pattern throughout the tissue. Such variation implies that even closely related TG species may diverge in their metabolic roles or interactions with immune cells in obese adipose tissue [20]. By pinpointing these distribution differences, we gain a clearer understanding of how lipid storage and turnover might shape local inflammatory processes and insulin sensitivity [10].

For triglycerides, the glycerol backbone carries three fatty acid chains. When we see total carbon:double bond numbers like TG(48:9), the composition might include combinations that sum to C48:9 (ex. 16:3 + 16:3 + 16:3). Certain combinations might produce more anti-inflammatory or pro-inflammatory signals upon partial hydrolysis (DGs, MGs), depending on the chain lengths and unsaturation. The existence of TG(48:9) in two distinct m/z peaks underscores how isomeric forms (different double-bond positions) can be resolved by MALDI IMS each possibly responding to different local enzymes or metabolic demands [6,7].

Local accumulation of TGs around necrotic adipocytes could reflect intense lipolysis, where adipocytes break down stored TGs, feeding FFAs into macrophages or forming partial glycerides en route. Conversely, some TG hotspots might represent sites of re-esterification, where excess FFA or MG is converted back into TG under certain inflammatory or metabolic cues. Such TG remodeling may influence whether macrophages adopt a pro-inflammatory phenotype (often fueled by saturated fatty acids like palmitate) versus a more oxidative or anti-inflammatory state (potentially linked to polyunsaturated fatty acids). Additionally, if certain TG species localize specifically to crown-like structures, they might serve as an “energy reservoir” for infiltrating macrophages or a source of secondary bioactive lipids like MG(20:3).

Correlating MG/TG “Hotspots” with Macrophage Activity

Correlating the MALDI IMS maps with known markers of macrophage infiltration suggests that MG/TG hotspots in crown-like structures could mirror active lipid metabolism, cytokine release, or phagocytic clearance of necrotic adipocytes [20]. By comparing MG(20:3) distribution with

transcriptomic data (upregulation of lipases or fatty acid oxidation genes), we can hypothesize that these localized lipids may act as both substrates and signals for macrophage-driven inflammation.

Eicosatrienoic-acid-containing MGs (MG(20:3)) may generate eicosanoid-like products that modulate macrophage cytokine production, while docosatrienoic or docosadienoic acids (MG(22:3), MG(22:2)) could feed into longer-chain mediator families still under investigation. Understanding whether these lipids ultimately skew macrophages toward pro- or anti-inflammatory phenotypes will require pairing spatial lipid data with functional readouts (cytokine levels, NF- κ B activation) in those same tissue regions [6,7].

In some regions, MG intensities overlapped partially with TG “rings,” implying potential in situ generation of MAGs from triglycerides under lipolytic conditions. This local interplay might feed into insulin resistance if pro-inflammatory macrophages are fueled by and simultaneously produce specific glycerolipids that disrupt adipocyte function. Understanding which MG/TG species drive or mitigate inflammation remains a key question.

Spatial Lipidomics and the Broader Context of Insulin Resistance

Overall, the spatial lipidomics findings emphasize that these rarely documented glycerolipids are not uniformly distributed throughout adipose tissue but show discrete patterns in inflamed vs. less inflamed zones [20]. For instance, MG(20:3) containing eicosatrienoic acid might have partial anti-inflammatory effects if it generates certain prostaglandins (ex. PGE1 from DGLA), whereas docosenoic or docosadienoic acids might influence membrane fluidity or oxidative stress responses. In contrast, some saturated or partially unsaturated TG species might exacerbate macrophage inflammation by releasing saturated FFAs upon hydrolysis [10].

Such patterns highlight the possibility that MG(20:3) or TG(46:9), for example, could be implicated in localized immune regulation, while others (ex. MG(22:1), TG(50:8)) may be more broadly involved in baseline lipid metabolism. Future mechanistic studies coupled with high-resolution imaging correlating macrophage markers will clarify whether the presence of these glycerolipids causally contributes to tissue inflammation, or if they simply reflect underlying adipocyte distress. These spatial observations lay the groundwork for multi-omic integration, tying lipid distribution data to transcriptomic signatures. By pinpointing the exact subregions where novel MG and TG species flourish,

we gain insight into the metabolic niches that may spur macrophage activation and perpetuate insulin resistance [6,7].

This targeted approach identifying lipid hotspots via IMS and then exploring local gene expression could ultimately reveal new therapeutic targets for curtailing obesity-induced inflammation and metabolic dysfunction. Altogether, the refined picture is that each MG or TG distinguished by its fatty acid composition can modulate or reflect macrophage-driven processes in obese adipose tissue [10]. Eicosatrienoic acid (MG(20:3)) and docosatrienoic acid (MG(22:3)), in particular, may serve as noteworthy intermediates with potential anti-inflammatory capacities, whereas certain saturated or partially unsaturated TG species (TG(48:9) isomers) could predispose adipose macrophages to heightened inflammatory responses [20]. Ongoing work linking these spatial distributions to functional assays will be critical in deciphering the true inflammatory or protective roles of these glycerolipids in metabolic disease [8].

Combining Bulk Lipidomics, Spatial IMS, and Transcriptomics in Obese Macrophages

Obesity-induced adipose inflammation involves macrophages infiltrating crown-like structures (CLSs), contributing to insulin resistance and metabolic dysfunction [11]. In this study, we examined how arachidonic acid (AA) and 2-arachidonoylglycerol (2-AG) influence macrophage lipid profiles using bulk lipidomics (LC–MS), supported by spatial lipidomics (MALDI IMS) to visualize lipid distributions in tissue sections, and a transcriptomic (RNA-seq) dataset highlighting gene-level regulatory shifts. This multi-layered approach provides a more comprehensive understanding of endocannabinoid- and eicosanoid-related processes in obese adipose tissue [12].

Principal Component Analysis (PCA) of Arachidonic Acid vs. 2-AG Treatments

The principal component analysis (PCA) of bulk lipidomic data showed a notable distinction between AA-treated samples and both vehicle and 2-AG groups, suggesting that AA drives more extensive glycerolipid remodeling [13]. Because arachidonic acid is a known precursor to pro-inflammatory mediators such as prostaglandins and leukotrienes [14], its higher leverage in the PCA aligns with the potential for AA to amplify inflammatory pathways in macrophages. In contrast, 2-AG–treated samples often clustered nearer to vehicle controls, implying that 2-AG may exert more subtle or targeted changes. These trends echo the notion that 2-AG, an endogenous endocannabinoid, can modulate rather than drastically elevate inflammatory signaling [15].

Heatmap Comparisons Highlight Distinct Lipid Classes Under AA or 2-AG

Hierarchical clustering heat maps provided a second lens on these global differences. Several AA-treated replicates grouped together, correlating with elevated intensities of certain phospholipids, fatty-acyl derivatives, or eicosanoid precursors. By contrast, 2-AG-treated replicates often formed subclusters that partially overlapped with vehicle, consistent with a less dramatic shift in overall lipid composition. In both treatments, some specific lipid classes such as phosphatidylethanolamine (PE), phosphatidylcholine (PC), or monoglycerides exhibited distinct up- or downregulation, indicating that macrophages adjust membrane-related and signaling lipids in response to these external stimuli [12,16]. The differential clustering underscores how the nature (pro-inflammatory vs. modulatory) of each treatment can selectively alter lipid metabolism.

Ceramides, Monoglycerides, and Phospholipids: Specific Shifts in Macrophages

To pinpoint which lipid classes are most altered, individual bar plots comparing ceramides, triglycerides, diglycerides, and monoglycerides revealed further details:

- **Ceramides and Triglycerides (TGs):** Some AA-treated samples showed upregulated TG species, possibly fueling local inflammatory loops if the released free fatty acids engage macrophage Toll-like receptors or other inflammatory pathways [11,17]. Meanwhile, ceramide accumulation has been linked to insulin resistance, hinting that the presence of specific saturated ceramides under AA could perpetuate metabolic stress [18].
- **Monoglycerides (MGs):** For example, MG(20:3) or MG(22:2) might reflect partial glyceride forms from AA or docosatrienoic acid. These intermediates can feed into eicosanoid-like signaling or endocannabinoid pathways [14]. If 2-AG-treated samples do not show as high a spike in these MG species, it suggests a relatively milder metabolic perturbation.
- **Phospholipids (PE, PC, PI, etc.):** Subtle but consistent changes in major membrane lipids reinforce the notion that the macrophage membrane environment adapts to external lipid cues, potentially altering membrane fluidity or receptor clustering relevant to inflammation [19].

Such class-specific differences fit with the broader suggestion that AA can strongly induce proinflammatory metabolite pools, whereas 2-AG exerts finer control or remains closer to baseline lipid profiles [15].

Localizing MG/TG “Hotspots” in Inflamed Adipose Tissue

Bulk analyses do not reveal where these lipids accumulate, a gap addressed by spatial lipidomics. MALDI IMS indicated that top monoglycerides and triglycerides (identified via LC–MS) localize preferentially near macrophage-rich crown-like structures. In AA-treated tissue, MG or TG hotspots often coincide with necrotic adipocytes, hinting at ongoing lipolysis or remodeling in pro-inflammatory niches [20]. In contrast, 2-AG tissues show fewer or smaller hotspots, implying a less regionally intense effect. When combined with the transcriptomic data revealing upregulation of eicosanoid biosynthesis genes (ex. COX2, PLA2) under AA, the spatial results suggest a synergy between local lipid accumulations and macrophage-driven inflammation.

Eicosanoid vs. Endocannabinoid Genes: Linking Lipid Species to Inflammatory Signaling

Many of the genes relevant to arachidonic acid metabolism for example COX2, ALOX5, and PTGS1 showed heightened expression in AA-treated macrophages, aligning with potential increased production of pro-inflammatory prostaglandins and leukotrienes [14]. In contrast, endocannabinoid-related enzymes like monoacylglycerol lipase (MGLL) or fatty acid amide hydrolase (FAAH) may be differentially regulated in 2-AG–treated cells [15,21]. This synergy is crucial: if the data show elevated levels of certain AA-based MGs (MG(20:4), MG(20:3)) in the lipidomics dataset and concurrent transcriptomic upregulation of pro-inflammatory genes, a functional link emerges. Conversely, 2-AG’s mild clustering in both heatmaps and PCA, coupled with a less pronounced up- or downregulation of inflammatory transcripts, supports a more immunomodulatory role for 2-AG, preventing the full-blown pro-inflammatory state that AA might induce [15].

Mediating Macrophage Inflammation through Lipidome Remodeling

Overall, these integrated findings reinforce the concept that arachidonic acid in macrophages can promote pro-inflammatory phenotypes via eicosanoid production and significant glycerolipid remodeling [11,14,17]. By contrast, 2-AG triggers only partial or modest changes in the global lipidome and in inflammatory gene expression, aligning with its endocannabinoid profile of modulating rather

than exacerbating inflammation [15,21]. Spatially, these alterations converge in CLSs and inflamed adipose regions where local lipid buildup, enzymatic turnover, and cytokine release coalesce to drive adipose tissue dysfunction [12]. Understanding these mechanisms is vital for appreciating how exogenous vs. endogenous lipid mediators shape macrophage activation, insulin resistance, and long-term metabolic health.

Bringing It All Together: Multi-Omics Integration Highlights AA and 2-AG Contrasts

In sum, the bulk lipidomics reveals global shifts; the spatial lipidomics localizes those shifts to inflamed tissue zones, and the transcriptomic data clarifies which pathways underlie these phenomena. This multi-omics integration underscores arachidonic acid as a potent inflammatory driver and 2-AG as a relatively mild modulator of macrophage lipid profiles paving the way for targeted studies linking these lipids to adipose inflammation in obesity [13,16].

Future Directions

Building on the insights gained from our multi-omics analyses, transcriptomics, spatial lipidomics, and bulk lipidomics, several lines of inquiry can further refine our understanding of lipid-driven macrophage activation and metabolic dysfunction. First, functional validation of key lipids–gene interactions represents a critical next step. Although our data suggest particular glycerolipids (arachidonic acid–derived monoglycerides) and genes (such as *COX2* or *ALOX5*) may play central roles in macrophage inflammation, targeted knockdowns (RNAi or CRISPR) and selective enzyme inhibition assays are needed to confirm whether these molecules actively drive or temper the pro-inflammatory state. Experimenting with lipase or cyclooxygenase inhibitors, for instance, could clarify how blocking specific lipid pathways alters cytokine outputs or insulin signaling in vitro.

A second avenue involves single-cell or spatial transcriptomic technologies, expanding on the synergy already established between MALDI IMS and bulk transcriptomics. By leveraging methods such as single-cell RNA-seq or spatial transcriptomics platforms (Visium, MERFISH), researchers could resolve the cell-to-cell heterogeneity among macrophages within crown-like structures. Such fine-grained analyses can reveal which macrophage subtypes accrue certain lipids and precisely how local eicosanoid or endocannabinoid signals shape inflammation at the cellular level. Additionally, stable isotope tracing and metabolic flux analysis could yield new perspectives on whether arachidonic acid (AA) versus 2-arachidonoylglycerol (2-AG) enter distinct metabolic fates, such as β -oxidation or storage. By

employing ^{13}C - or ^2H -labeled forms of these lipids in macrophage cultures, we could pinpoint exactly where label incorporation leads to eicosanoids or endocannabinoids, determining which pathways predominate under particular inflammatory stimuli.

Further in vivo validation, using rodent models of diet-induced obesity, would help translate these discoveries. Specifically, testing how pharmacological agents such as monoacylglycerol lipase (MGLL) inhibitors or CB2 receptor modulators affect adipose inflammation and insulin resistance could demonstrate whether lipid–gene relationships observed here hold therapeutic potential. Lastly, a longitudinal time-course design would capture the dynamic evolution of macrophage polarization and lipid shifts at early, intermediate, and advanced obesity stages. This could ascertain whether 2-AG exerts a protective influence during initial disease phases, or if AA progressively accumulates to promote more severe pro-inflammatory cascades. By exploring these integrative and mechanistic extensions, future work will help define precisely how endocannabinoid versus eicosanoid pathways orchestrate macrophage function and metabolic health, ultimately illuminating new targets for controlling obesity-related inflammation.

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Appendix: Abbreviations and Concentration Notes

General / Methods

- **ATCC: American Type Culture Collection**
A global repository and distributor of authenticated cell lines and microorganisms.
- **Bodipy: Boron-dipyrromethene**
A fluorescent dye used to visualize lipid droplets via confocal or fluorescence microscopy.
- **DMEM: Dulbecco's Modified Eagle Medium**
A commonly used cell culture medium, often supplemented with glucose, FBS, and antibiotics.
- **FBS: Fetal Bovine Serum**
A serum supplement that provides nutrients and growth factors in cell culture media.
- **PBS: Phosphate-Buffered Saline**
A buffer solution often used for washing cells or tissues.

Macrophage & Adipose Biology

- **CLSs: Crown-Like Structures**
Macrophage-rich aggregates around necrotic or hypertrophic adipocytes in obese adipose tissue.
- **Dmpk: Dystrophia Myotonica-Protein Kinase**
A gene commonly associated with muscle function but also implicated in macrophage responses.
- **Cpt1a: Carnitine Palmitoyltransferase 1A**
Enzyme regulating mitochondrial β -oxidation of fatty acids, crucial in metabolic reprogramming of macrophages.
- **GPCR: G-Protein–Coupled Receptor**
A large family of membrane receptors that can sense molecules outside the cell (ex. monoglycerides) to trigger intracellular responses.
- **TLR: Toll-Like Receptor**
Receptors recognizing pathogen or damage-associated molecules; can drive pro-inflammatory signaling in macrophages.
- **NF- κ B: Nuclear Factor Kappa-B**
A key transcription factor in inflammatory pathways, activated by various lipid and cytokine

signals.

Concentration / Treatments

- **High-Dose (5 mM):** The 5 mM concentration is referred to throughout the thesis as the high dose used for monoglyceride, OA, or PA treatments, known to elicit robust changes in macrophage gene expression and lipid metabolism.
- **Low-Dose (500 μ M):** The 500 μ M concentration is the low dose, capturing subtler, more physiologically relevant shifts in transcriptomic profiles. Comparing these doses clarifies dose-dependency in macrophage polarization and metabolic outcomes.

Lipids and Related Terms

- **AA: Arachidonic Acid**
A 20-carbon polyunsaturated fatty acid; precursor for eicosanoids (prostaglandins, leukotrienes).
- **2-AG: 2-Arachidonoylglycerol**
An endogenous monoglyceride functioning as an endocannabinoid, impacting inflammation and metabolism.
- **OA: Oleic Acid**
A monounsaturated fatty acid (18:1). Often studied for its roles in metabolic regulation and macrophage lipid uptake.
- **PA: Palmitic Acid**
A saturated fatty acid (16:0) frequently linked to pro-inflammatory states in adipose macrophages.
- **MAG (Monoglyceride):**
A glycerol molecule esterified to one fatty acid (ex. MG(20:3), MG(22:3)). Can act as signaling lipids or metabolic intermediates.
- **MGLL: Monoacylglycerol Lipase**
An enzyme that hydrolyzes monoglycerides to free fatty acids (FFAs) and glycerol, influencing inflammation and energy metabolism.
- **FFA: Free Fatty Acid**
A fatty acid not bound to a glycerol backbone; can serve as fuel or engage inflammatory pathways.

- **DG (Diglyceride):**
A glycerol molecule with two fatty acid chains.
- **TG (Triglyceride):**
A glycerol molecule with three fatty acid chains; the main storage form of fat in adipocytes.
- **DGLA: Dihomo- γ -linolenic Acid**
An n-6 fatty acid that can produce less pro-inflammatory mediators than arachidonic acid.
- **FAAH: Fatty Acid Amide Hydrolase**
Breaks down certain endocannabinoids (ex. anandamide).

Eicosanoid / Endocannabinoid

- **COX2: Cyclooxygenase-2**
An enzyme converting arachidonic acid to prostaglandins, often upregulated in inflammation.
- **ALOX5: 5-Lipoxygenase**
Converts arachidonic acid to leukotrienes, another pro-inflammatory branch of eicosanoid production.
- **PTGS1: Prostaglandin-Endoperoxide Synthase 1**
Also known as COX1, involved in prostaglandin biosynthesis.

Multi-Omic Techniques

- **MALDI IMS: Matrix-Assisted Laser Desorption/Ionization Imaging Mass Spectrometry**
Used for spatial lipidomics, mapping lipid species directly in tissue sections.
- **LC-MS: Liquid Chromatography-Mass Spectrometry**
A “bulk” approach to quantify overall lipid composition, including free fatty acids, MAGs, DAGs, TGs, and specialized lipids (AA, 2-AG).
- **RNA-seq: RNA Sequencing**
A next-generation sequencing method for global gene expression profiling in macrophages.